Lecture 7-2: Quantum mechanics and atomic structure 1

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

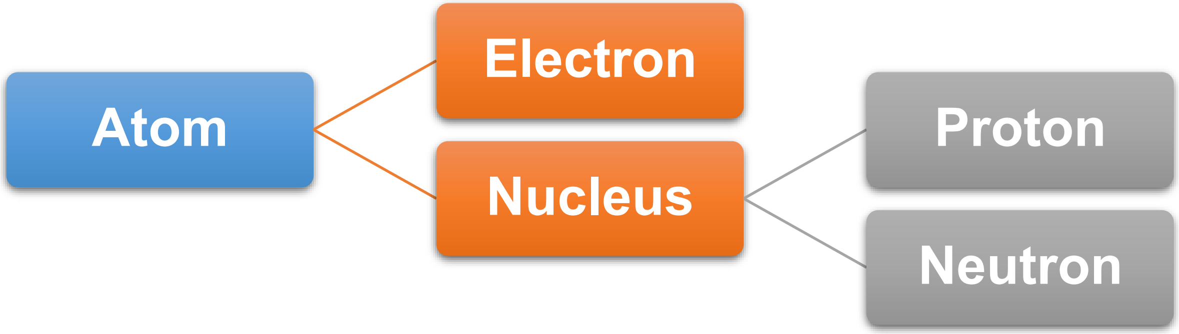
2025/10/24

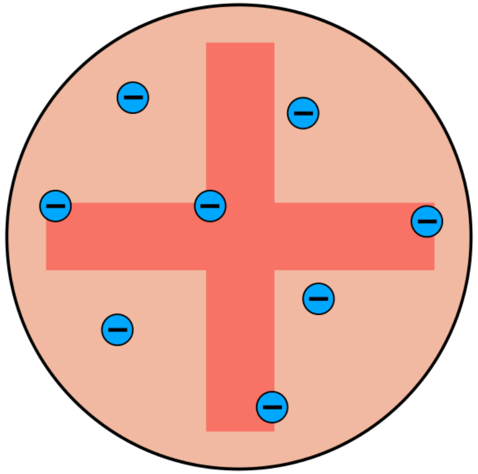
Major points of today's lecture

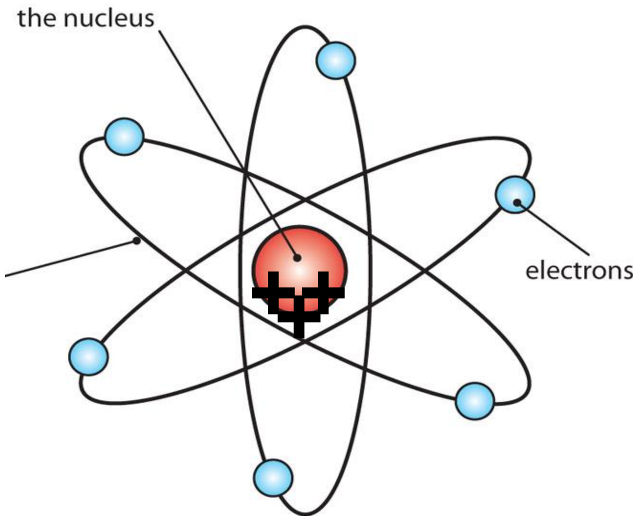
- **Quantum mechanics and atomic structure**

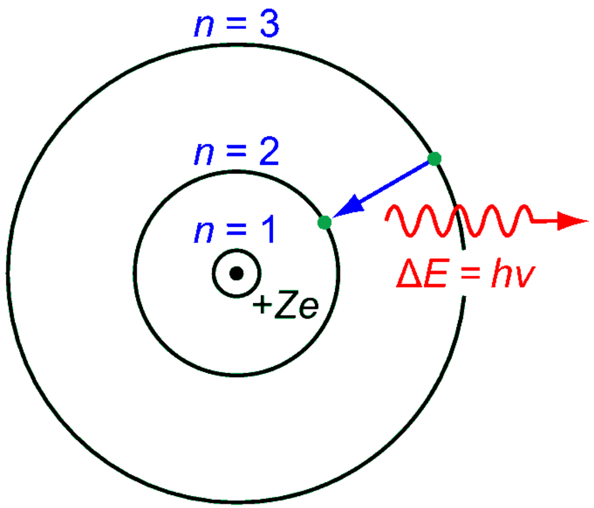
1. The hydrogen atom
2. Shell model for many-electron atoms
3. Aufbau principle and electron configuration
4. Shells and the periodic table: photoelectron spectroscopy
5. Periodic properties and electronic structure

Summary of previous atomic structure









But the Bohr model doesn’t work for the He atom at all!

The hydrogen atom: Schrödinger equation

the Schrödinger equation (time-independent)

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]\psi = E\psi$$

动能算符　　势能算符

time independent Schrödinger equation for H atom

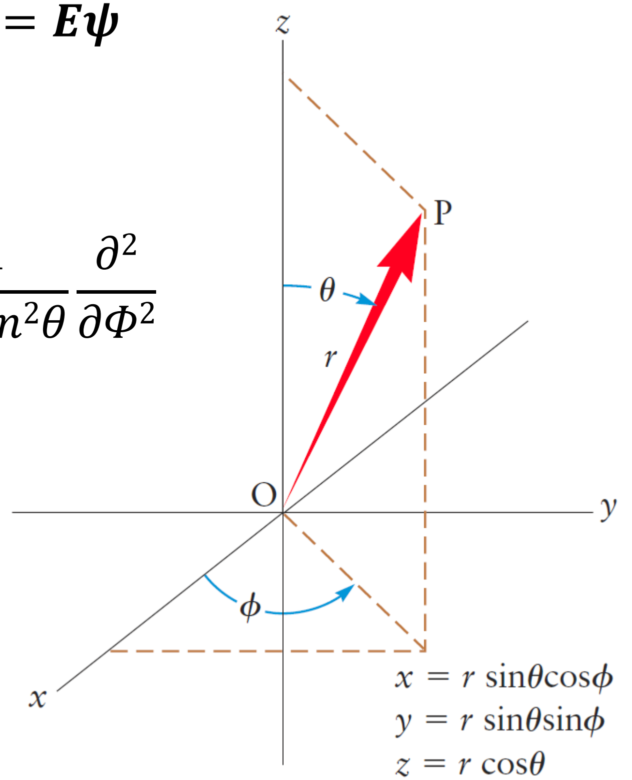
$$\left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right]\psi = E\psi$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Cartesian coordinates (笛卡尔坐标)

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}r^2\frac{\partial}{\partial r} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\Phi^2}$$

Spherical coordinates (球坐标)



$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$
$$x^2 + y^2 + z^2 = r^2$$

Solve the Schrödinger equation　$\psi_{n\ell m}(r, \theta, \phi)$

The hydrogen atom: wave functions

$$\psi_{n\ell m}(r,\theta,\phi) = \boxed{R_{n\ell}(r)}\boxed{Y_{\ell m}(\theta,\phi)}$$

radial part angular part

For each quantum state (n, ℓ, m), solution of the Schrödinger equation provides a wave function:

TABLE 5.2 Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$	Radial Part $R_{n\ell}(r)$
$\ell = 0$: $Y_s = \left(\frac{1}{4\pi}\right)^{1/2}$	$R_{1s} = 2\left(\frac{Z}{a_0}\right)^{3/2} \exp(-\sigma)$
	$R_{2s} = \frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2}(2-\sigma)\exp(-\sigma/2)$
	$R_{3s} = \frac{2}{81\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2}(27-18\sigma+2\sigma^2)\exp(-\sigma/3)$
$\ell = 1$: $Y_{p_x} = \left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\cos\phi$	$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma\exp(-\sigma/2)$
$Y_{p_y} = \left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\sin\phi$	$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}(6\sigma-\sigma^2)\exp(-\sigma/3)$
$Y_{p_z} = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$	
$\ell = 2$: $Y_{d_{z^2}} = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$	$R_{3d} = \frac{4}{81\sqrt{30}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma^2\exp(-\sigma/3)$
$Y_{d_{xz}} = \left(\frac{15}{4\pi}\right)^{1/2}\sin\theta\cos\theta\cos\phi$	
$Y_{d_{yz}} = \left(\frac{15}{4\pi}\right)^{1/2}\sin\theta\cos\theta\sin\phi$	
$Y_{d_{xy}} = \left(\frac{15}{16\pi}\right)^{1/2}\sin^2\theta\sin 2\phi$	
$Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi}\right)^{1/2}\sin^2\theta\cos 2\phi$	

$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

The hydrogen atom: energy levels

$$\left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}\right]\psi = E\psi$$

Solutions of the Schrödinger equation for the one-electron atom exist only for particular values of the energy

$$E = E_n = -\frac{Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} \qquad n = 1, 2, 3, \ldots$$

$$E_n = -\frac{Z^2}{n^2} \quad \text{(rydberg)} \qquad n = 1, 2, 3, \ldots$$

The integer n (principal quantum number, 主量子数) indexes the individual energy levels. These are identical to the energy levels predicted by the Bohr theory.

Quantization arises because of the physical boundary conditions imposed on the solutions to the Schrodinger equation (需要满足边界条件) rather than from making arbitrary assumptions about the angular momentum.

The hydrogen atom: quantum numbers

The Schrödinger equation **also quantizes L^2**, the square magnitude of the angular momentum (L), as well as **L_z**, the projection of the angular momentum along the z-axis (L_z).

$$L^2 = \ell(\ell+1)\frac{h^2}{4\pi^2} \qquad \ell = 0, 1, \ldots, n-1$$

The **angular momentum quantum number ℓ (角量子数)** , may take on any integral value from 0 to n-1

$$L_z = m\frac{h}{2\pi} \qquad m = -\ell,\ -\ell+1, \ldots, 0, \ldots, \ell-1,\ \ell$$

The **magnetic quantum number m (磁量子数)**, may take on any integral value from -ℓ to ℓ

Its value governs the behavior of the atom in an external magnetic field.

The hydrogen atom: quantum numbers

$n = 1, 2, 3, \ldots$ $\ell = 0, 1, \ldots, n-1$ $m = -\ell, -\ell + 1, \ldots, 0, \ldots, \ell - 1, \ell$

When n=1. Only one set of quantum states: n=1, l=0, m=0

When n=2. Four sets of quantum states: (2, 0, 0); (2, 1, -1); (2, 1, 0); (2, 1, 1)

When n=n. 1+3+5+…+(2n-1)=n^2 sets of quantum states

- **Each set (n, ℓ, m) identifies a specific quantum state of the atom in which the electron has energy equal to E_n.**
- **When n > 1, a total of n^2 specific quantum states correspond to the single energy level E_n; consequently, this set of states is said to be degenerate (简并).**

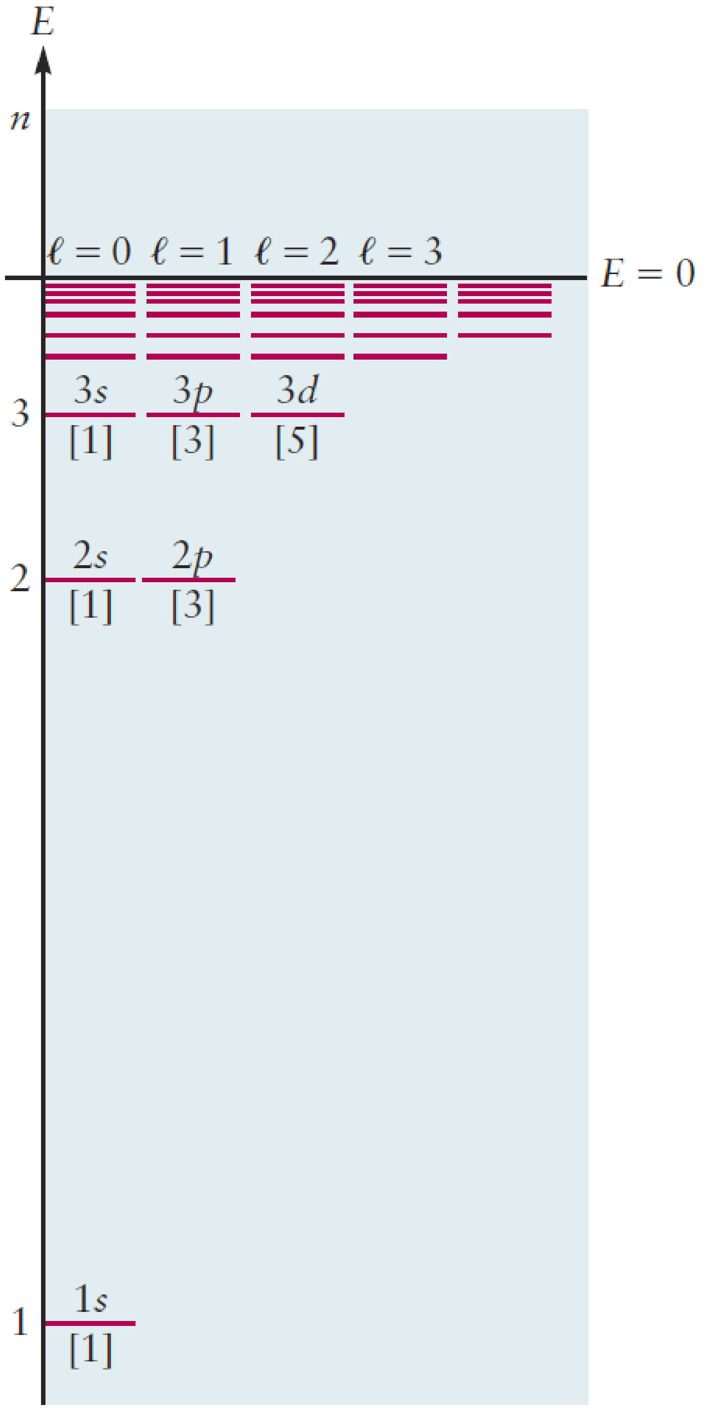
The hydrogen atom: quantum numbers

It is conventional to label specific states by replacing the angular momentum quantum number with a letter

$\ell = 0$ with s, $\ell = 1$ with p, $\ell = 2$ with d, $\ell = 3$ with f, $\ell = 4$ with g

TABLE 5.1 Allowed Values of Quantum Numbers for One-Electron Atoms

n	**1**	**2**		**3**		
ℓ	0	0	1	0	1	2
m	0	0	−1, 0, +1	0	−1, 0, +1	−2, −1, 0, +1, +2
Number of degenerate states for each ℓ	1	1	3	1	3	5
Number of degenerate states for each n	1	4		9		



Give the names of all the orbitals with $n = 4$, and state how many m values correspond to each type of orbital.

Solution

The quantum number ℓ may range from 0 to $n - 1$; thus, its allowed values in this case are 0, 1, 2, and 3. The labels for the groups of orbitals are then:

$\ell = 0$	4s	$\ell = 2$	4d
$\ell = 1$	4p	$\ell = 3$	4f

The quantum number m ranges from $-\ell$ to $+\ell$; thus, the number of m values is $2\ell + 1$. This gives one 4s orbital, three 4p orbitals, five 4d orbitals, and seven 4f orbitals for a total of $16 = 4^2 = n^2$ orbitals with $n = 4$. They all have the same energy, but they differ in shape.

The hydrogen atom: wave functions

For each quantum state (n, ℓ, m), solution of the Schrödinger equation provides a wave function:

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

radial part, angular part

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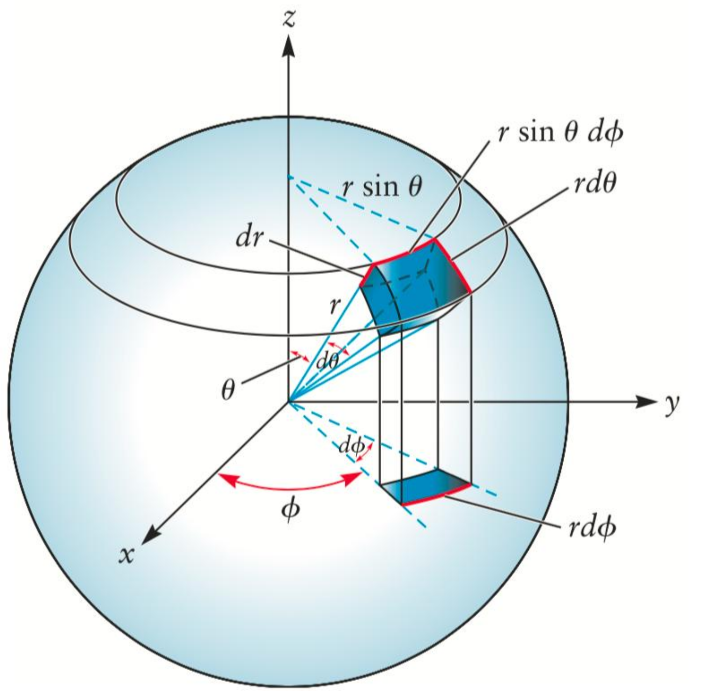
$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

The hydrogen atom: wave functions

probability of locating the electron

$$(\psi_{n\ell m})^2 dV = [R_{n\ell}(r)]^2[Y_{\ell m}(\theta,\phi)]^2 dV$$

dV, located at the position (r, θ, ϕ) when it is known that the atom is in the state n, ℓ, m.



Cartesian coordinates (笛卡尔坐标)

to spherical coordinates (球坐标)

$$dV = dxdydz = r^2 sin\theta \mathrm{d}r\mathrm{d}\theta\mathrm{d}\phi$$

The hydrogen atom: orbital

A wave function $\psi_{n\ell m}(r, \theta, \phi)$ for a one-electron atom in the state (n, ℓ, m) is called an orbital.

When the one-electron atom is in state (n, ℓ, m), it is conventional to say the electron is "in an (n, ℓ, m) orbital."

When an electron has energy, total angular momentum, and z component of angular momentum values corresponding to the quantum numbers n, ℓ, m, the probability density of finding the electron at the point (r, θ, ϕ) is given by $\psi^2{}_{n\ell m}(r, \theta, \phi)$.

Orbital is not some sort of "region in space" inside which the electron is confined.

vs. Bohr model for H atom

The hydrogen atom: orbital

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

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$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

1s, 2s, 3s

The hydrogen atom: s orbitals

s ORBITALS: corresponding to $\psi_{n\ell m}(r, \theta, \phi)$ with ℓ=0 (m = 0 as well).

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

$$\ell = 0 \left\{ Y_s = \left(\frac{1}{4\pi}\right)^{1/2} \right.$$

$$R_{1s} = 2\left(\frac{Z}{a_0}\right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} (2-\sigma)\exp(-\sigma/2)$$

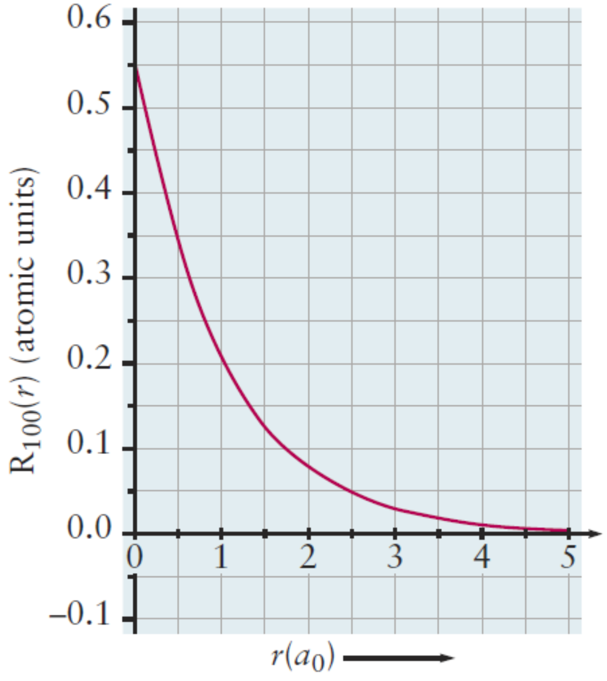
$$R_{3s} = \frac{2}{81\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2} (27 - 18\sigma + 2\sigma^2)\exp(-\sigma/3)$$

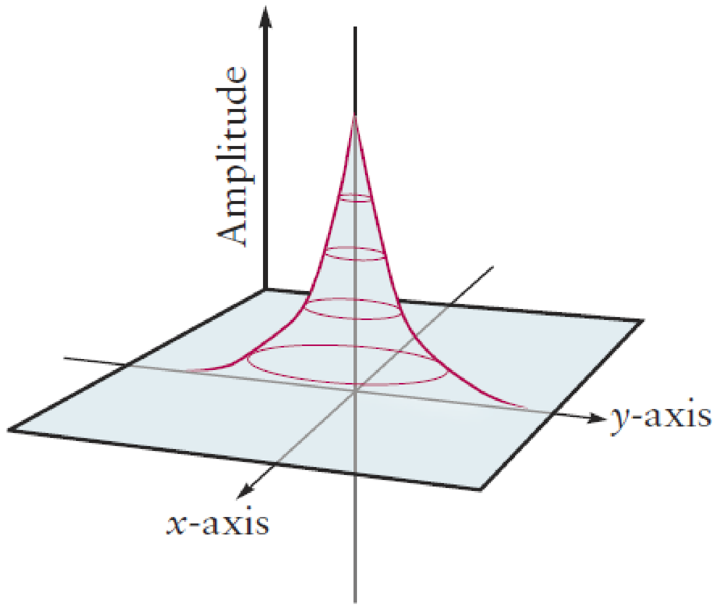
$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

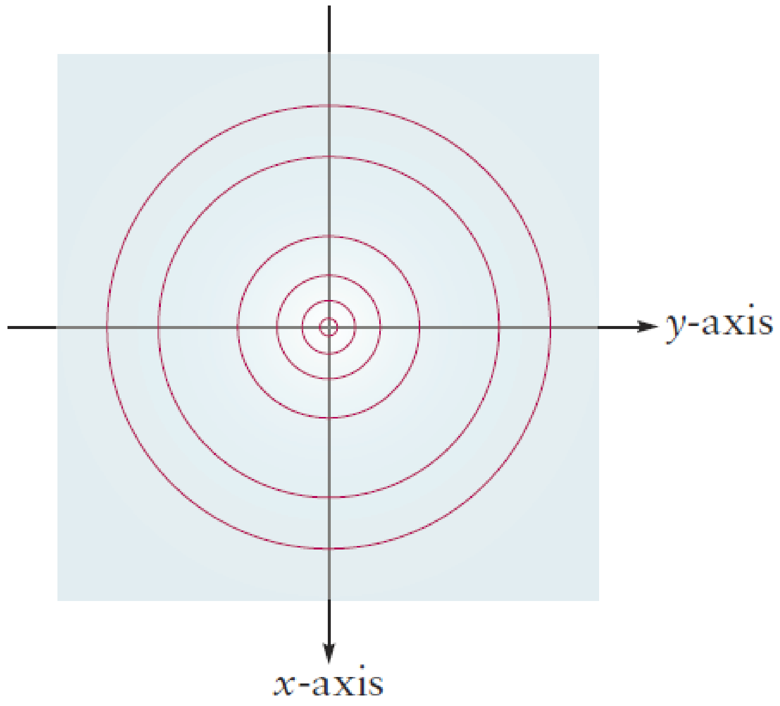
$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right) \qquad \psi_{2s} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \left(2 - \frac{r}{a_0}\right) \exp\left(-\frac{r}{2a_0}\right)$$

The hydrogen atom: 1s orbitals

$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right)$$

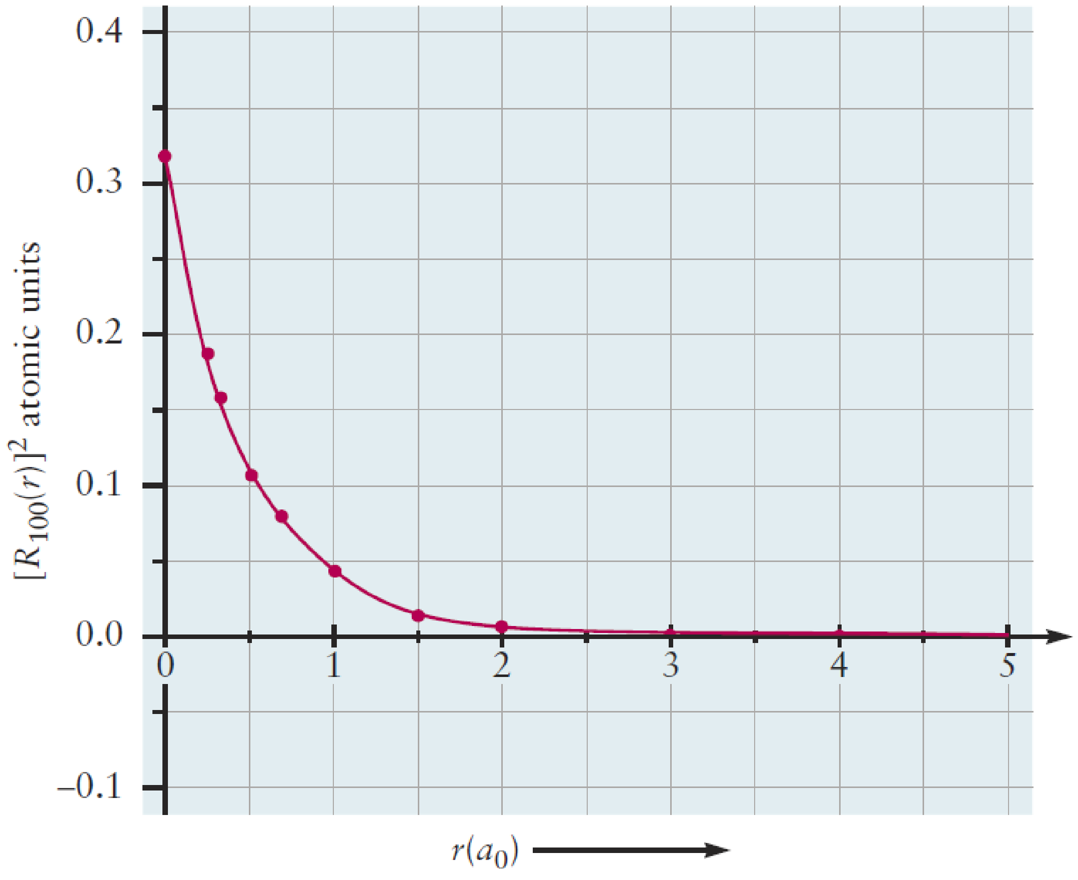


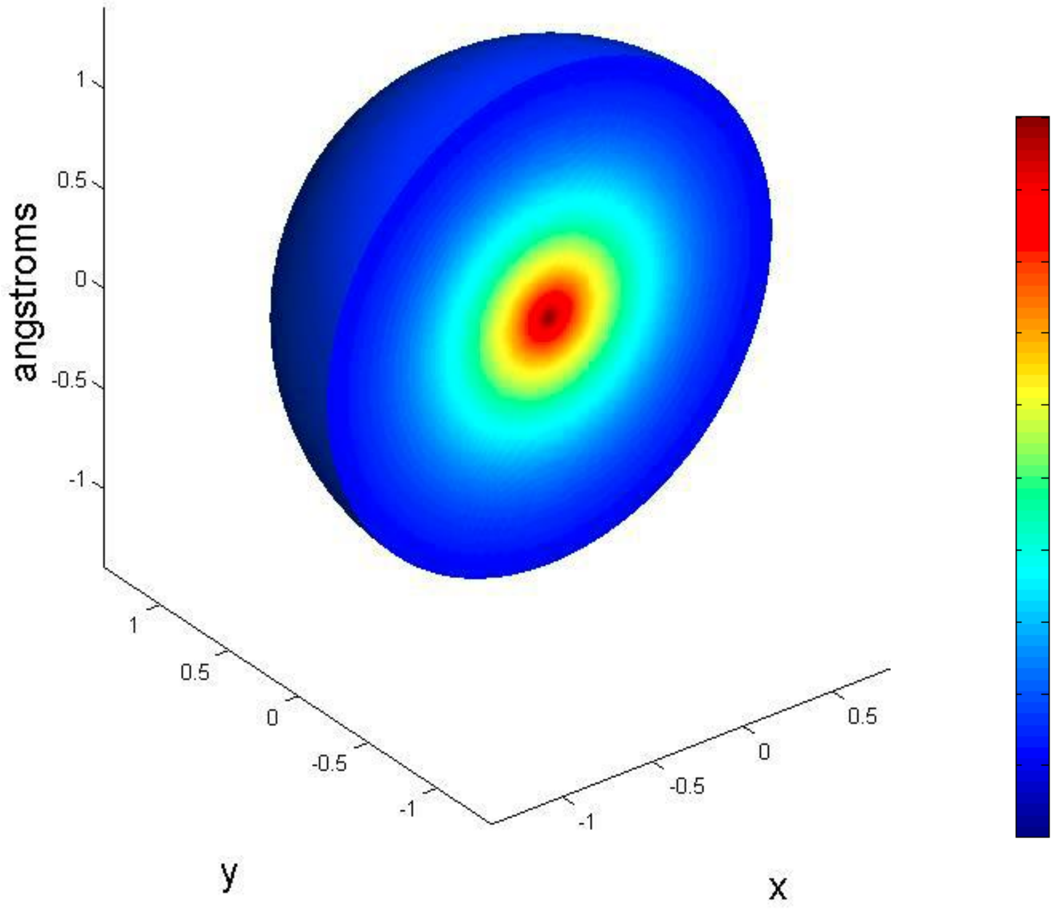




The hydrogen atom: 1s orbitals

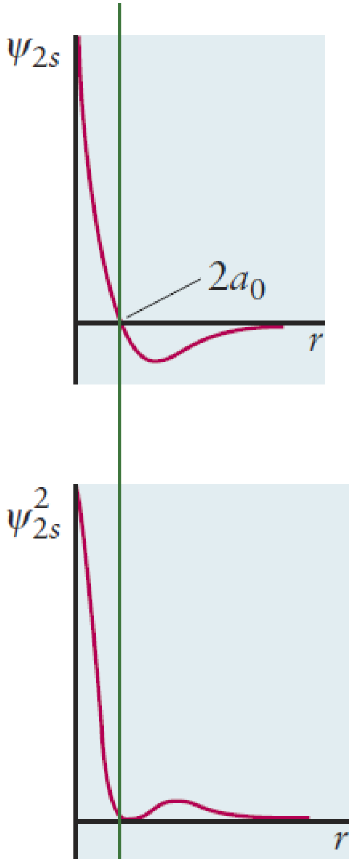
$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right)$$

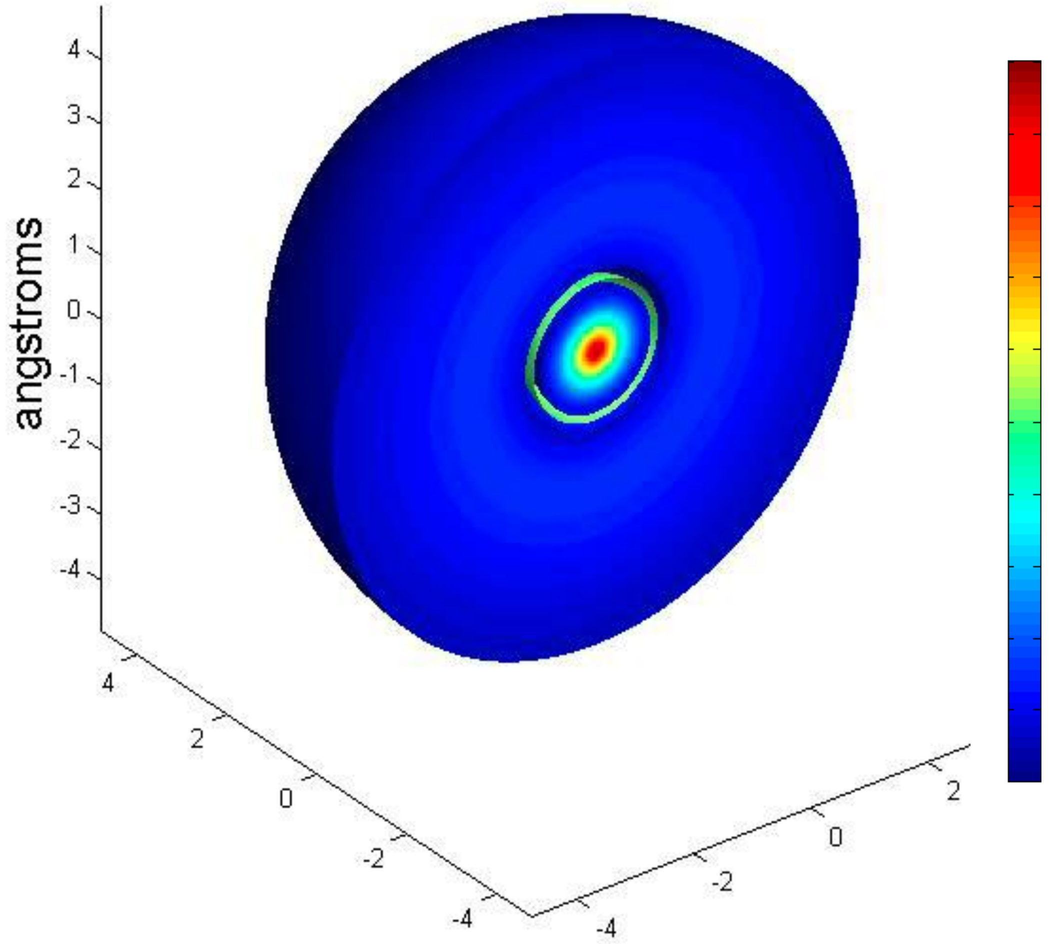




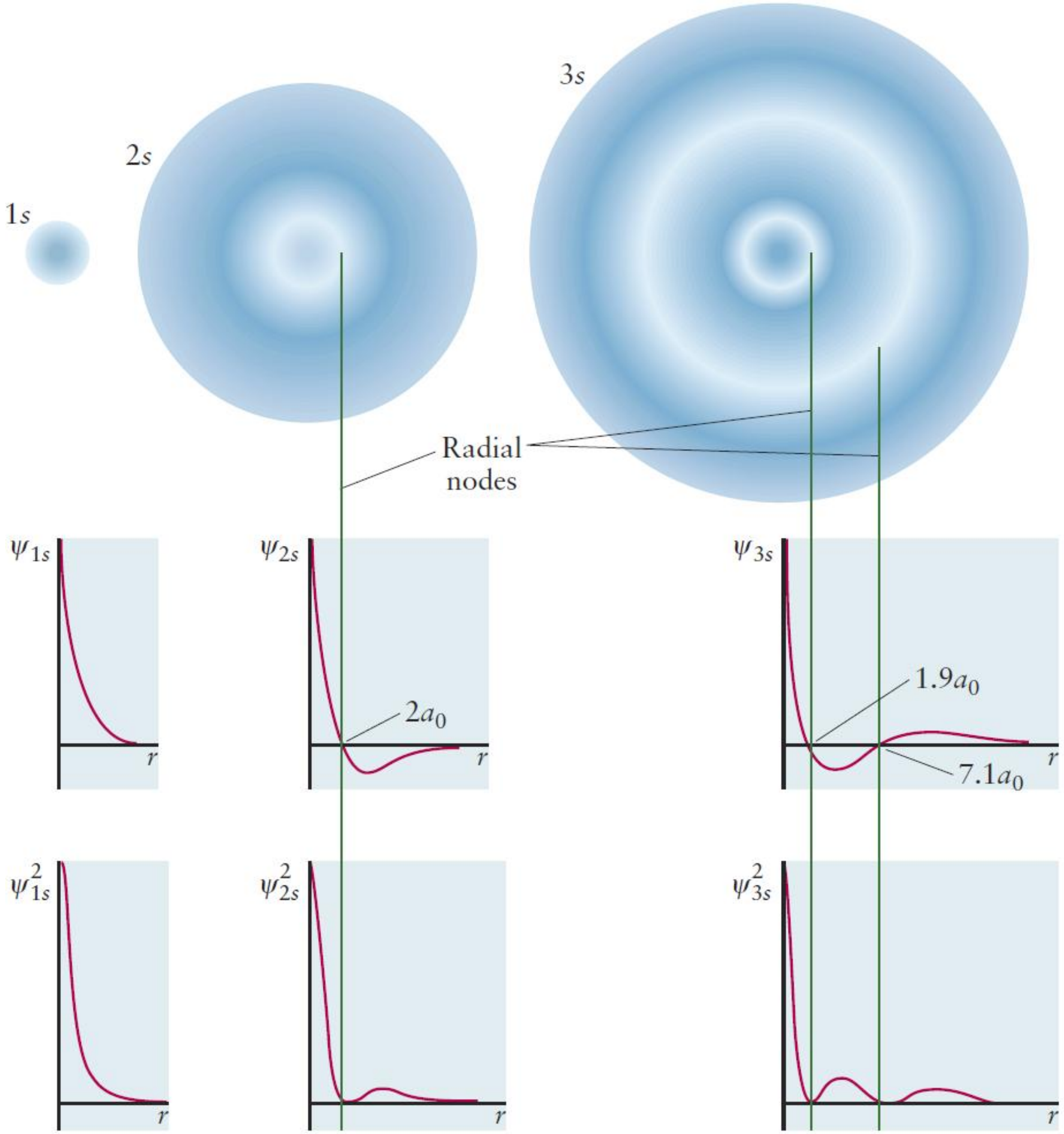
The hydrogen atom: 2s orbitals

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \left(2 - \frac{r}{a_0}\right) \exp\left(-\frac{r}{2a_0}\right)$$





The hydrogen atom: s orbitals

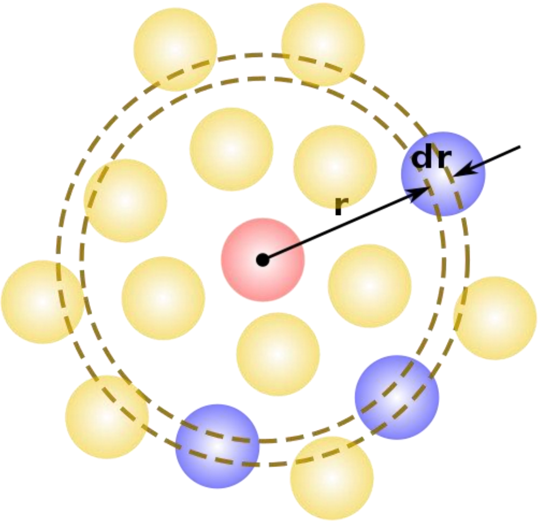


$$R_{1s} = 2\left(\frac{Z}{a_0}\right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} (2-\sigma)\exp(-\sigma/2)$$

$$R_{3s} = \frac{2}{81\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2} (27-18\sigma+2\sigma^2)\exp(-\sigma/3)$$

The hydrogen atom: s orbitals



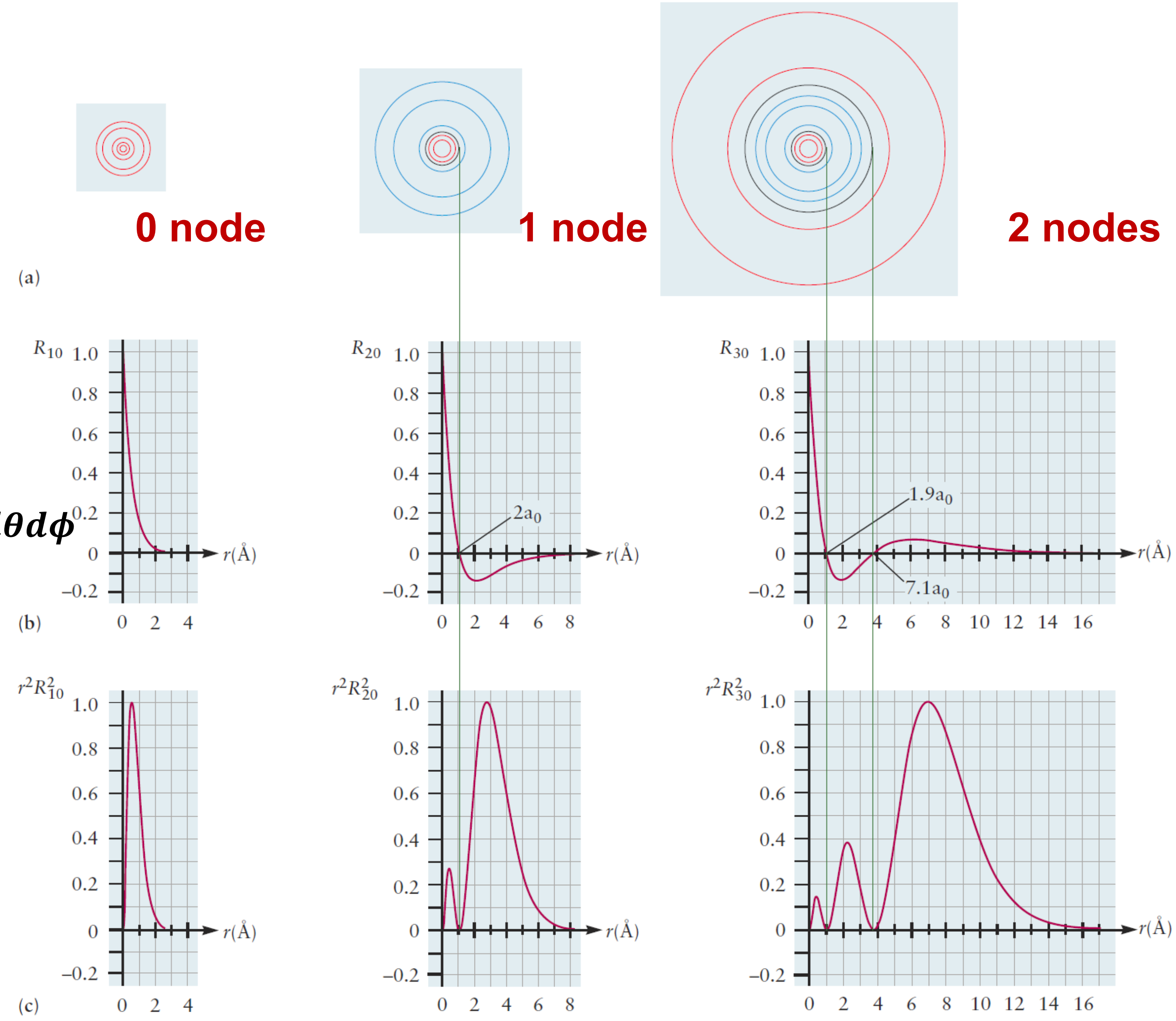
$$\int_V |\psi_s|^2 dV = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi}\int_{r=0}^{r=rx} |R_s(r)Y_s(\theta,\phi)|^2 r^2 sin\theta dr d\theta d\phi$$

$$= \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi}\frac{1}{4\pi}\, sin\theta d\theta d\phi \int_{r=0}^{r=rx} |R_s(r)|^2 r^2 dr$$

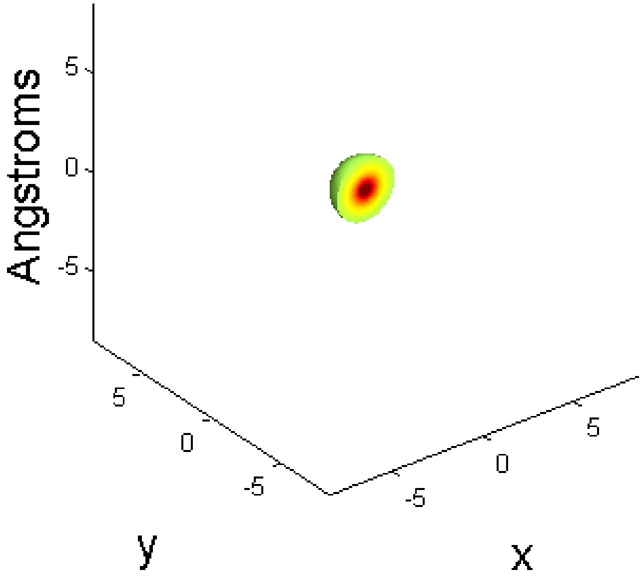
$$= (2\pi)(2)(\frac{1}{4\pi})\int_{r=0}^{r=rx} |R_s(r)|^2 r^2 dr$$

$$= \int_{r=0}^{r=rx} |R_s(r)|^2 r^2 dr$$

$$D = |R_s(r)|^2 r^2$$

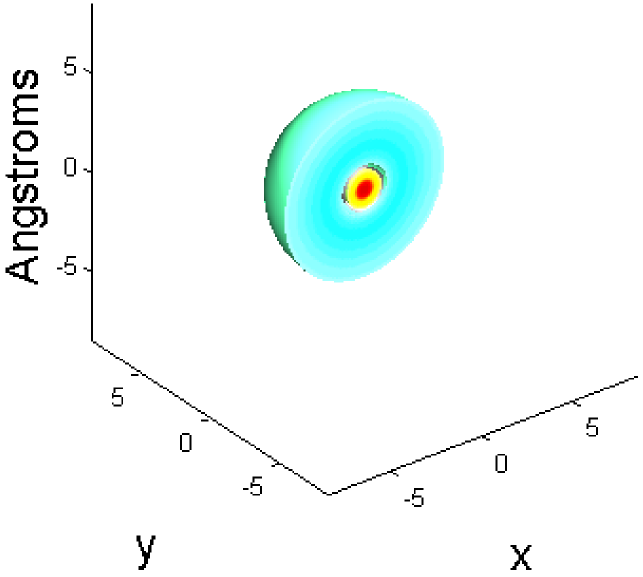


The hydrogen atom: Comparison of 1s, 2s, 3s Orbitals



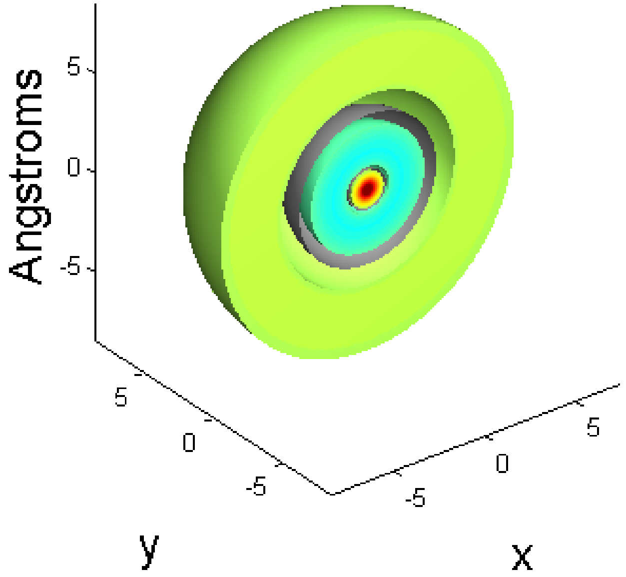
1*s*

n = 1, l = 0.



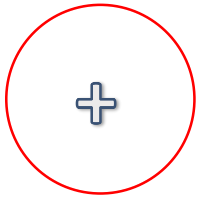
2*s*

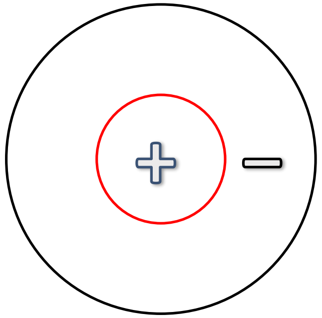
n = 2, l = 0.

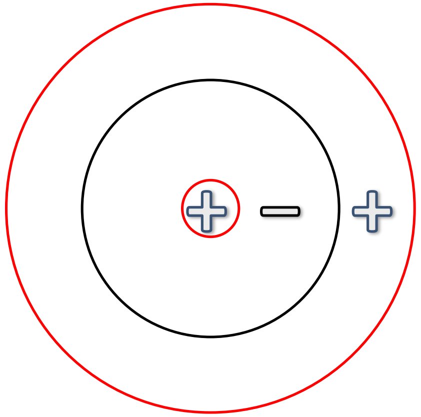


3*s*

n = 3, l = 0.







The hydrogen atom: orbital

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

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$\ell = 1$: $Y_{p_x} = \left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\cos\phi$	
$Y_{p_y} = \left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\sin\phi$	$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma\exp(-\sigma/2)$
$Y_{p_z} = \left(\frac{3}{4\pi}\right)^{1/2} \cos\theta$	$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}(6\sigma - \sigma^2)\exp(-\sigma/3)$
$\ell = 2$: $Y_{d_{z^2}} = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$	
$Y_{d_{xz}} = \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\cos\phi$	
$Y_{d_{yz}} = \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\sin\phi$	$R_{3d} = \frac{4}{81\sqrt{30}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma^2\exp(-\sigma/3)$
$Y_{d_{xy}} = \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta\sin 2\phi$	
$Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta\cos 2\phi$	

$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

$2p_x$, $2p_y$, $2p_z$

$3p_x$, $3p_y$, $3p_z$

(why there is no 1p?)

The hydrogen atom: p orbitals

p ORBITALS: ℓ=1, m=-1, 0, 1

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

$$\ell = 1 \begin{cases} Y_{p_x} = \left(\dfrac{3}{4\pi}\right)^{1/2} \sin\theta\cos\phi \\ Y_{p_y} = \left(\dfrac{3}{4\pi}\right)^{1/2} \sin\theta\sin\phi \\ Y_{p_z} = \left(\dfrac{3}{4\pi}\right)^{1/2} \cos\theta \end{cases}$$

$$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

Orbitals with angular quantum numbers different from 0 are not spherically symmetric

The hydrogen atom: p orbitals

Angular wave function

1 angular nodes

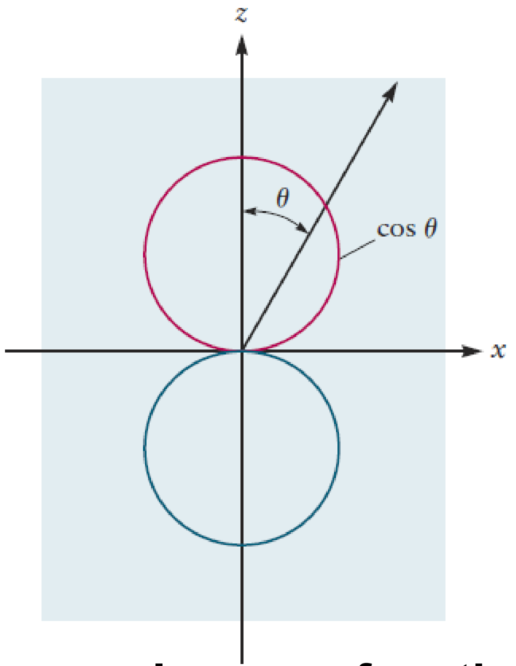
$$\ell = 1 \begin{cases} Y_{p_x} = \left(\dfrac{3}{4\pi}\right)^{1/2} \sin\theta\cos\phi \\ Y_{p_y} = \left(\dfrac{3}{4\pi}\right)^{1/2} \sin\theta\sin\phi \\ Y_{p_z} = \left(\dfrac{3}{4\pi}\right)^{1/2} \cos\theta \end{cases}$$

Radial wave function

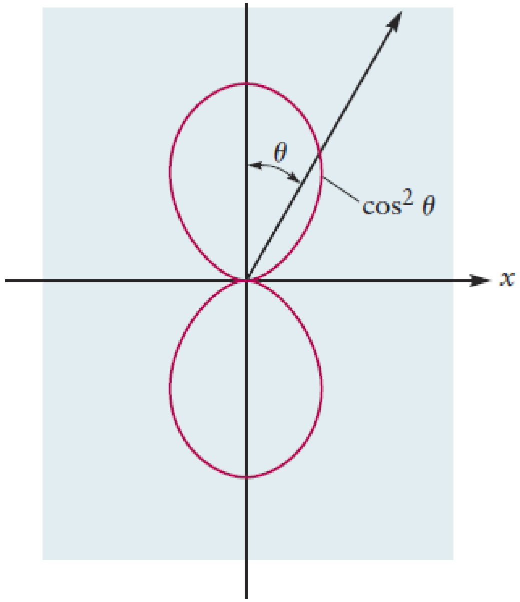
$$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} (6\sigma - \sigma^2)\exp(-\sigma/3)$$

n-ℓ-1 radial nodes



The angular wave function for the p_z orbital. The p_x and p_y orbitals are oriented along the x- and y-axis, respectively.



The square of the angular wave function for the pz orbital.

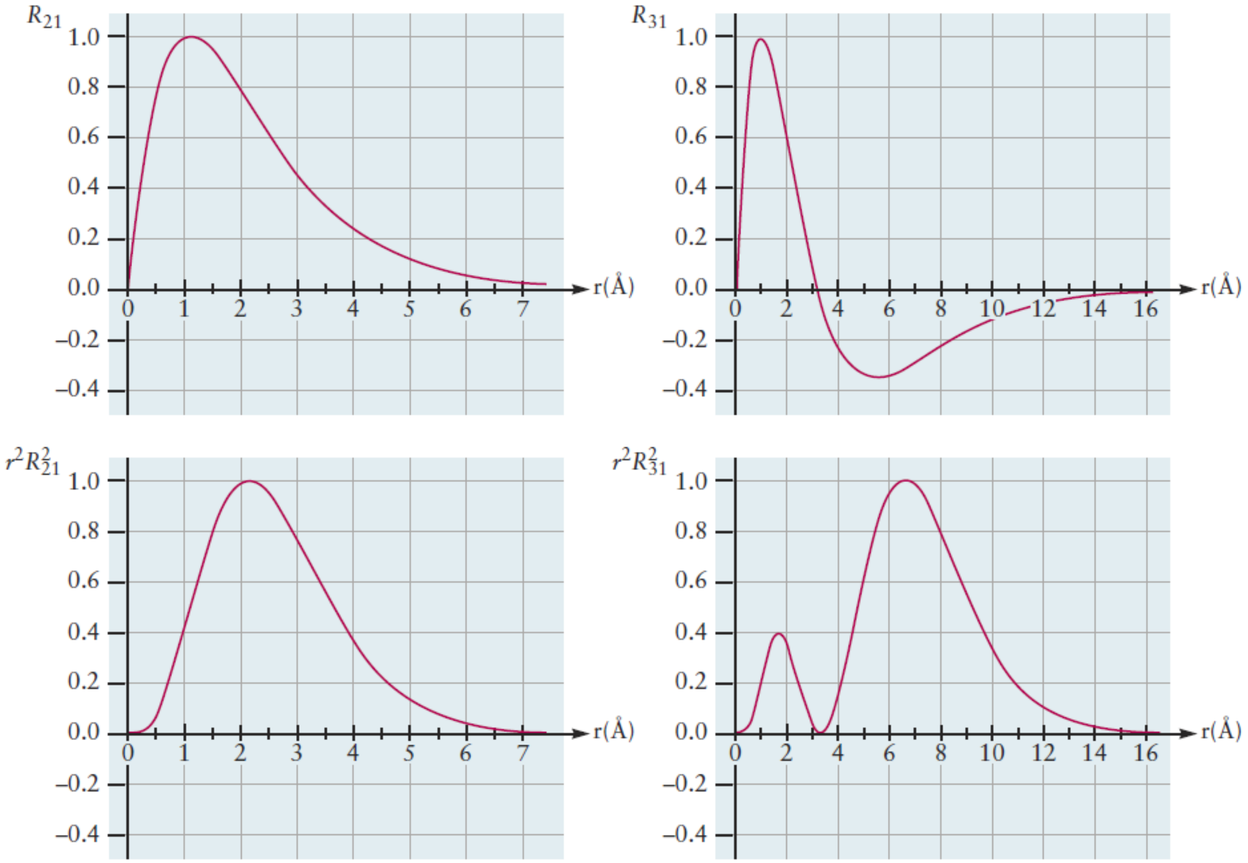
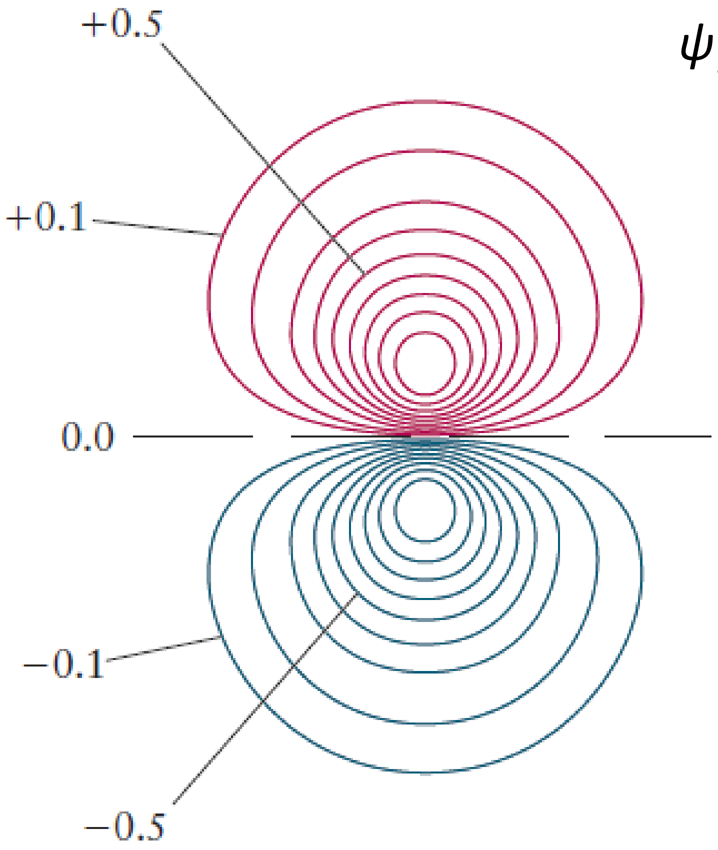


FIGURE 5.10 Radial wave functions $R_{n\ell}$ for *np* orbitals and the corresponding radial probability densities $r^2R_{n\ell}^2$.

The hydrogen atom: $2p_z$ orbitals

FIGURE 5.11 Contour plot for the amplitude in the p_z orbital for the hydrogen atom. This plot lies in the *x-z* plane. The *z*-axis (not shown) would be vertical in this figure, and the *x*-axis (not shown) would be horizontal. The lobe with positive phase is shown in red, and the lobe with negative phase in blue. The *x-y* nodal plane is shown as a dashed black line. Compare with Figure 5.9a.

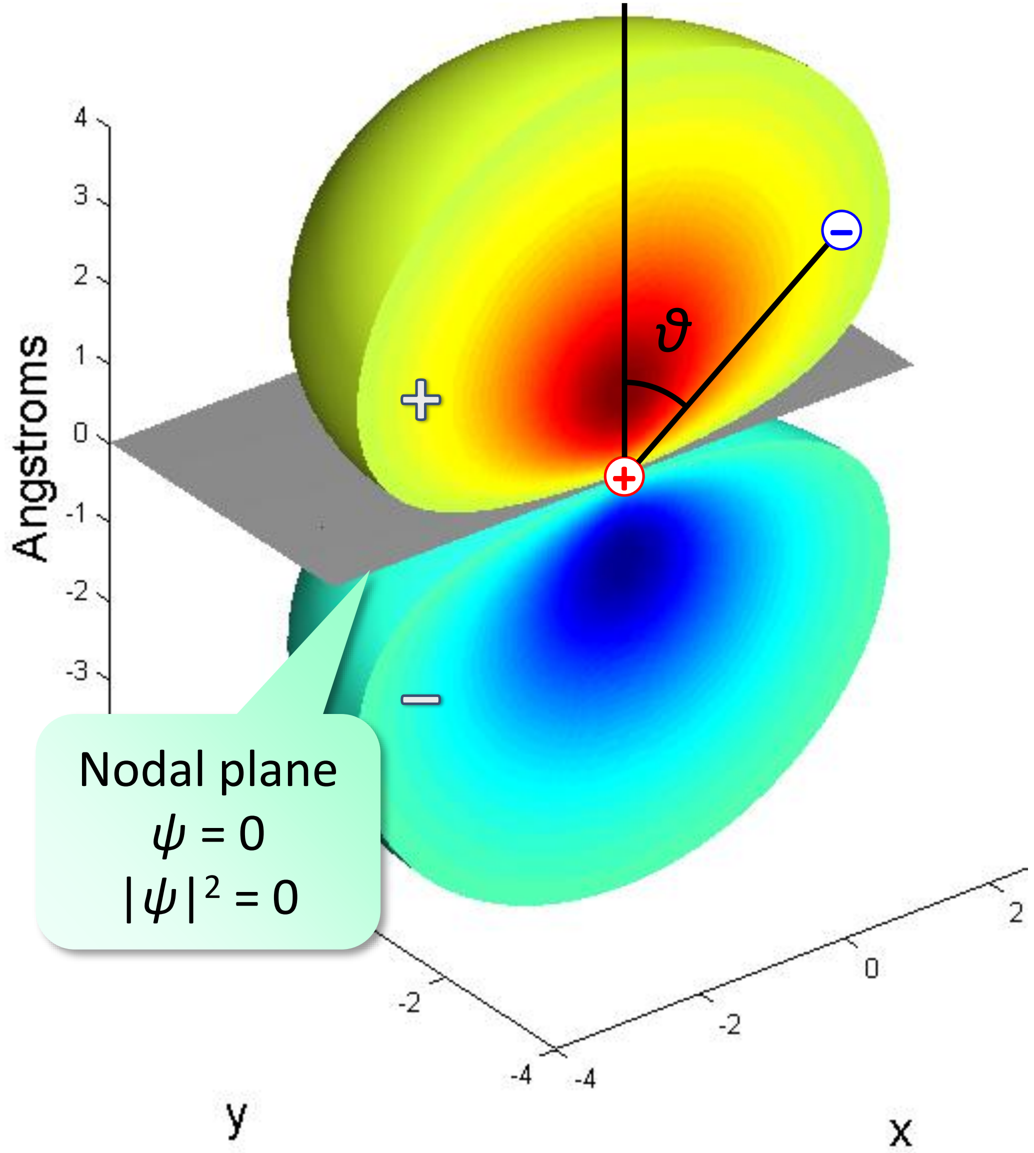


$$\psi_{2pz} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \underbrace{\frac{r}{a_0} \exp\left(-\frac{r}{2a_0}\right)}_{\text{Radial part, } R_{21}} \cdot \underbrace{\cos\vartheta}_{\text{Augular part, } Y_{pz}}$$

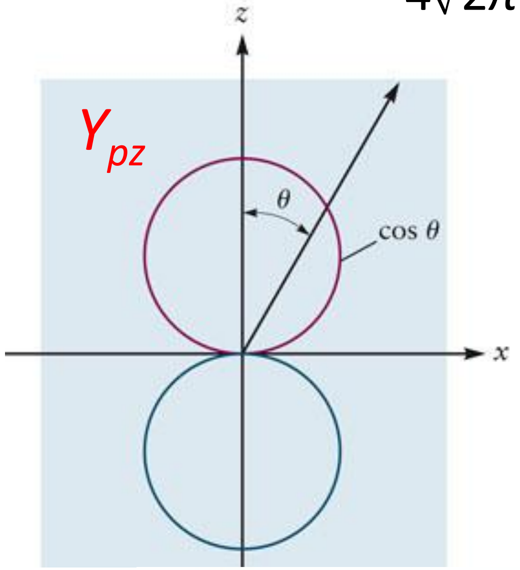
This plane is a nodal plane or, more generally, an angular node

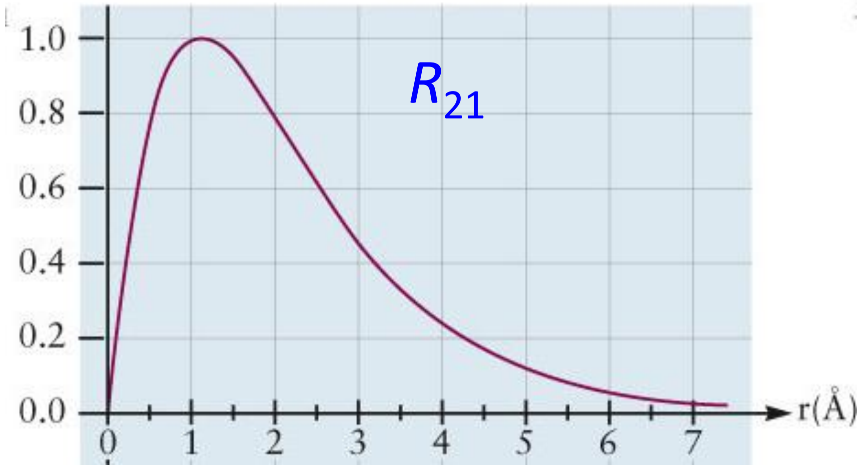
The wave function changes sign across it

The hydrogen atom: $2p_z$ orbitals

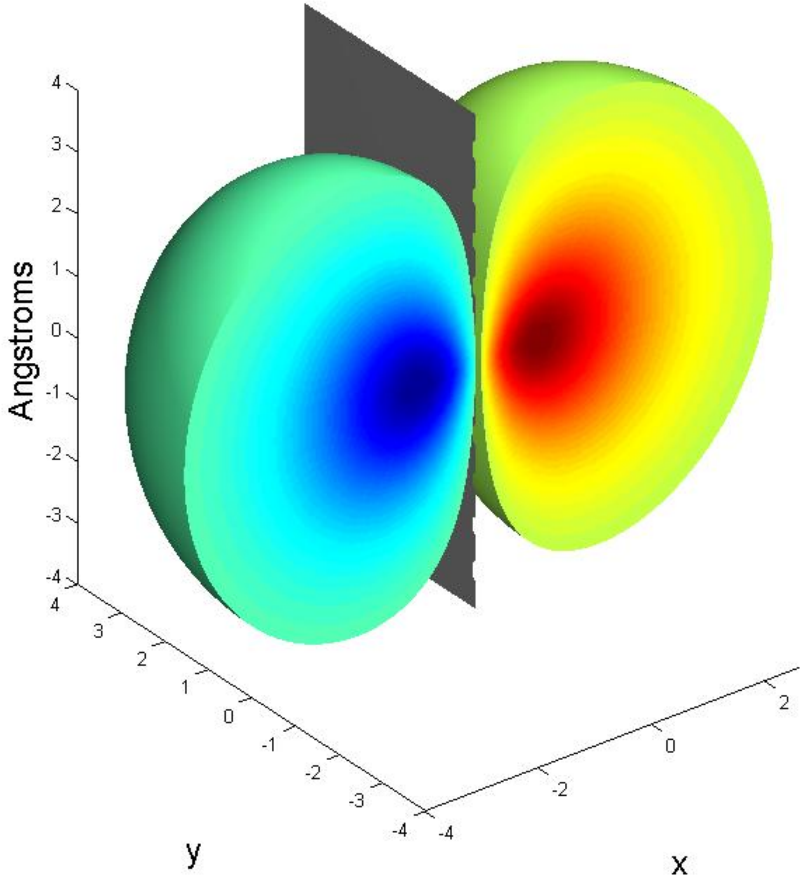


$$\psi_{2pz} = \frac{1}{4\sqrt{2\pi}}\, a_0^{-3/2} \cdot \underbrace{\frac{r}{a_0}\exp\left(-\frac{r}{2a_0}\right)}_{\text{Radial part, } R_{21}} \cdot \underbrace{\cos\vartheta}_{\text{Augular part, } Y_{pz}}$$

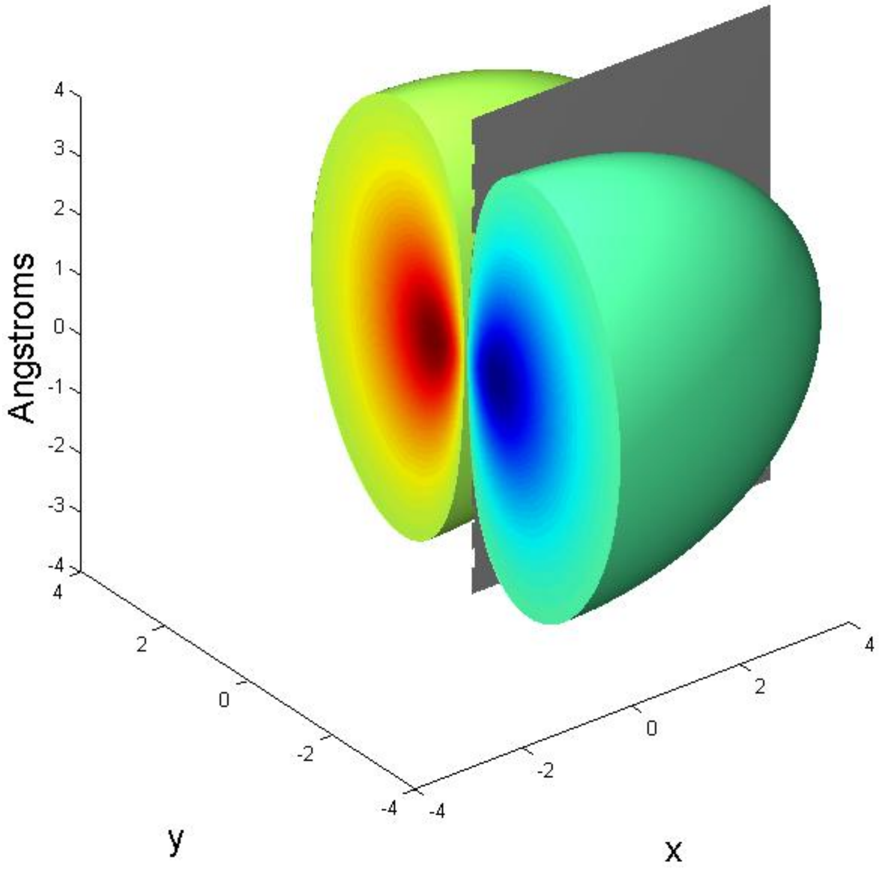




The hydrogen atom: $2p_x$ and $2p_y$ orbitals

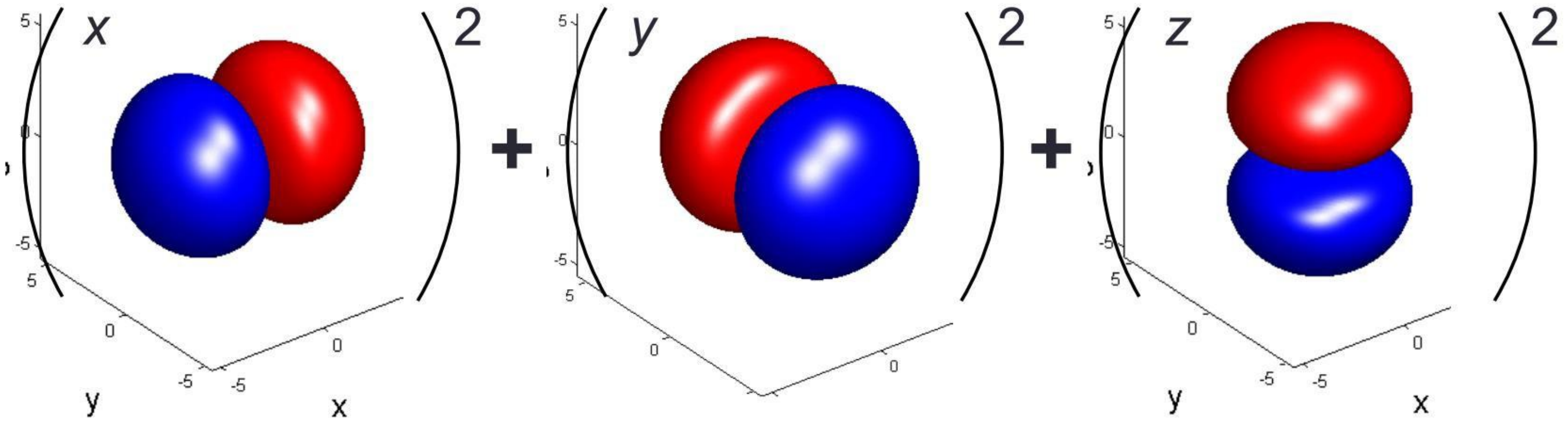


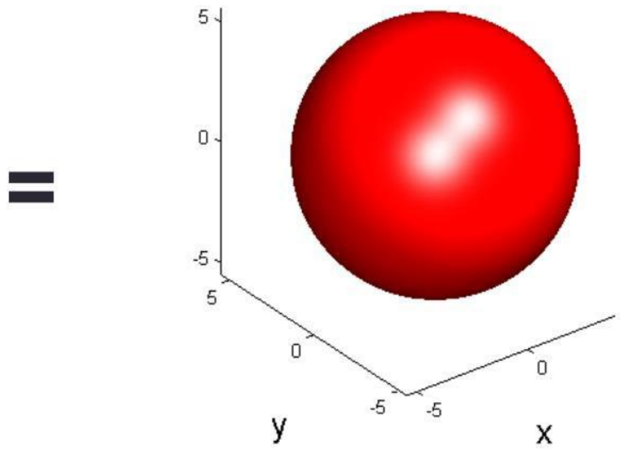
$2p_x$



$2p_y$

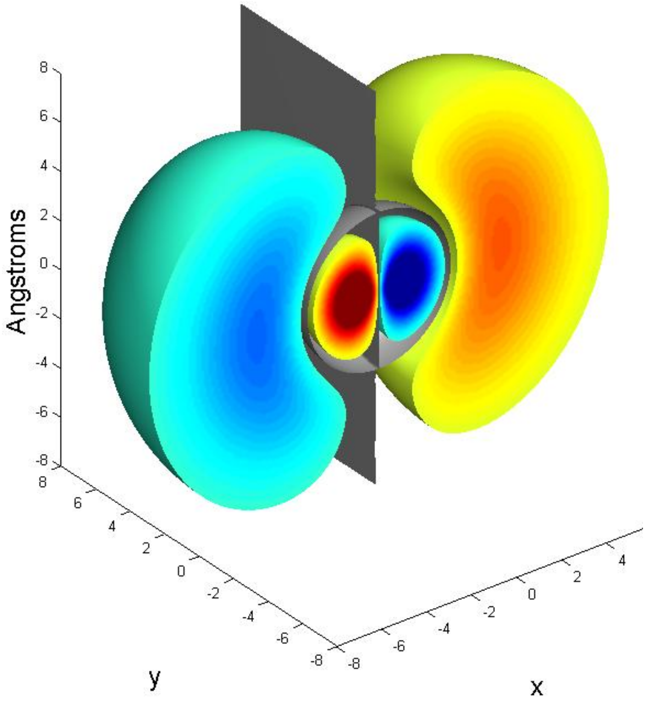
The hydrogen atom: Overlap of p Orbitals



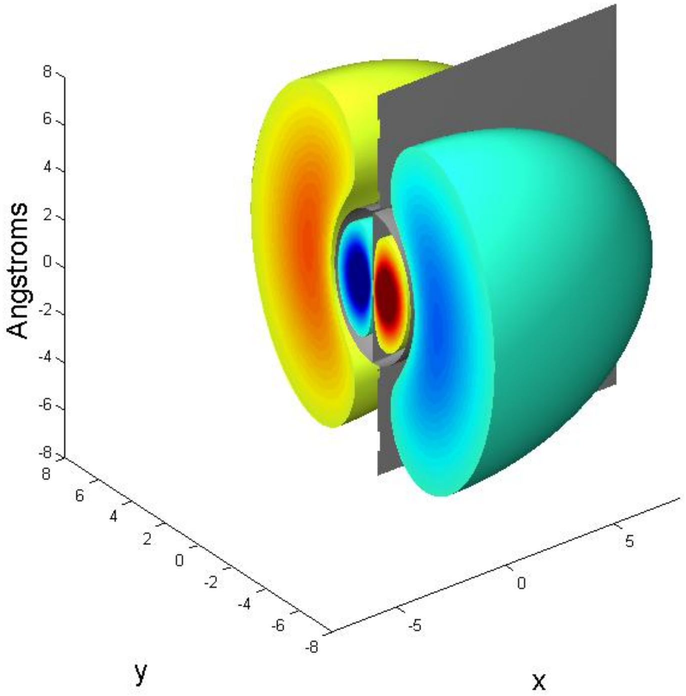


The three *p* orbitals occupy
separate parts of the space.

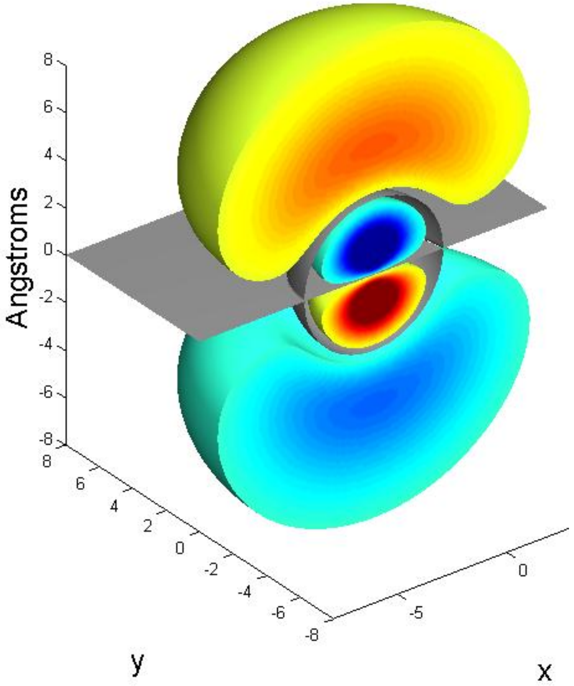
The hydrogen atom: 3p orbitals



$3p_x$



$3p_y$



$3p_z$

The hydrogen atom: orbital

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$$

TABLE 5.2 Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$	Radial Part $R_{n\ell}(r)$
	$R_{1s} = 2\left(\frac{Z}{a_0}\right)^{3/2} \exp(-\sigma)$
$\ell = 0$: $Y_s = \left(\frac{1}{4\pi}\right)^{1/2}$	$R_{2s} = \frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2}(2 - \sigma)\exp(-\sigma/2)$
	$R_{3s} = \frac{2}{81\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2}(27 - 18\sigma + 2\sigma^2)\exp(-\sigma/3)$
$\ell = 1$: $Y_{p_x} = \left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\cos\phi$	$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma\exp(-\sigma/2)$
$Y_{p_y} = \left(\frac{3}{4\pi}\right)^{1/2}\sin\theta\sin\phi$	$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}(6\sigma - \sigma^2)\exp(-\sigma/3)$
$Y_{p_z} = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$	
$\ell = 2$: $Y_{d_{z^2}} = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$	
$Y_{d_{xz}} = \left(\frac{15}{4\pi}\right)^{1/2}\sin\theta\cos\theta\cos\phi$	
$Y_{d_{yz}} = \left(\frac{15}{4\pi}\right)^{1/2}\sin\theta\cos\theta\sin\phi$	$R_{3d} = \frac{4}{81\sqrt{30}}\left(\frac{Z}{a_0}\right)^{3/2}\sigma^2\exp(-\sigma/3)$
$Y_{d_{xy}} = \left(\frac{15}{16\pi}\right)^{1/2}\sin^2\theta\sin 2\phi$	
$Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi}\right)^{1/2}\sin^2\theta\cos 2\phi$	

$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

3d…

The hydrogen atom: d orbitals

d ORBITALS: ℓ=2, m=-2, -1, 0, 1, 2

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

$$\ell = 2 \begin{cases} Y_{d_{z^2}} = \left(\dfrac{5}{16\pi}\right)^{1/2} (3\cos^2\theta - 1) \\ Y_{d_{xz}} = \left(\dfrac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\cos\phi \\ Y_{d_{yz}} = \left(\dfrac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\sin\phi \\ Y_{d_{xy}} = \left(\dfrac{15}{16\pi}\right)^{1/2} \sin^2\theta\sin 2\phi \\ Y_{d_{x^2-y^2}} = \left(\dfrac{15}{16\pi}\right)^{1/2} \sin^2\theta\cos 2\phi \end{cases}$$

$$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0}\right)^{3/2} \sigma^2 \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0} \qquad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10}\ \text{m}$$

The hydrogen atom: d orbitals

d orbital: 2 angular nodes, n-3 radial nodes. n-1 total nodes

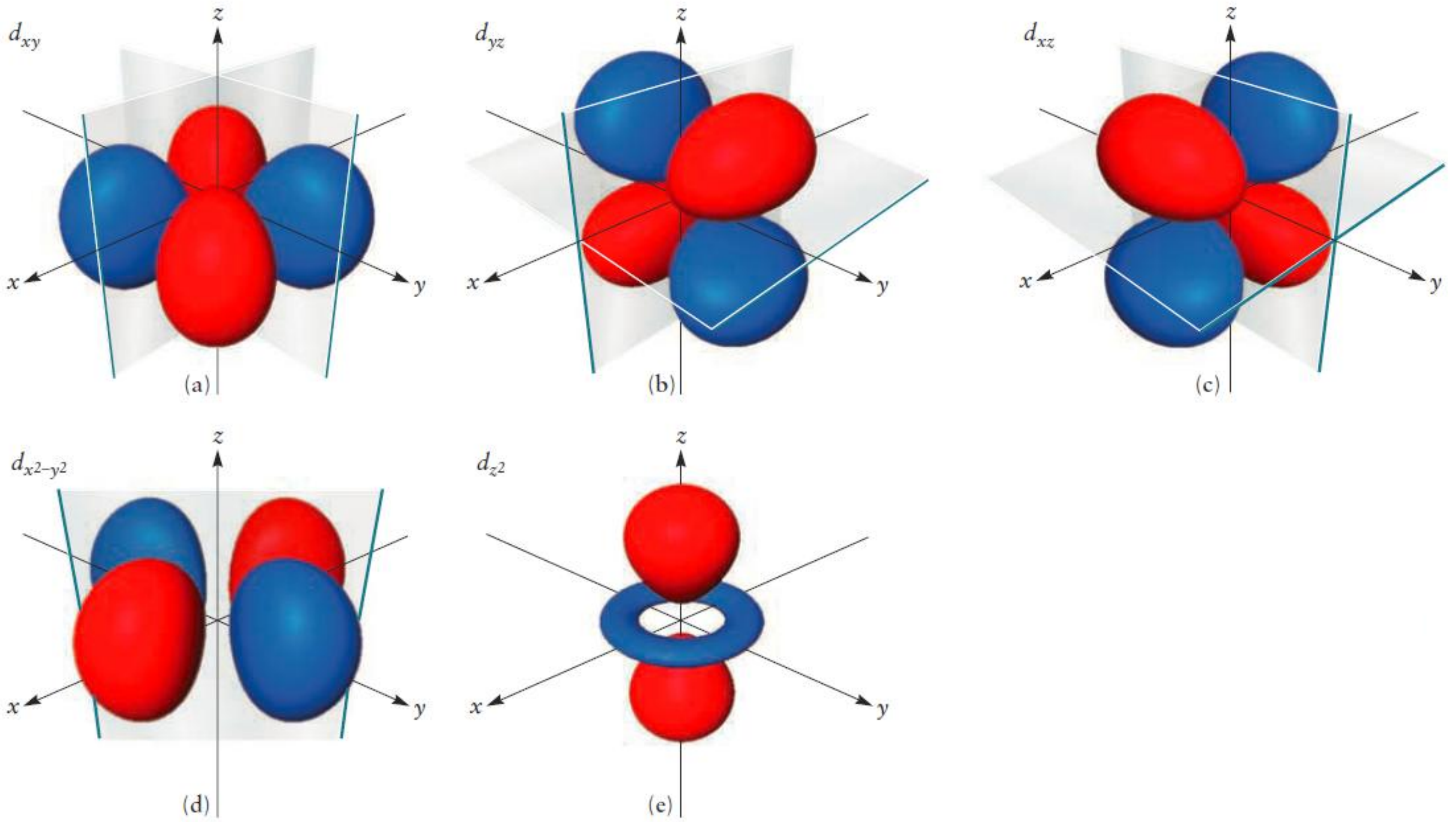
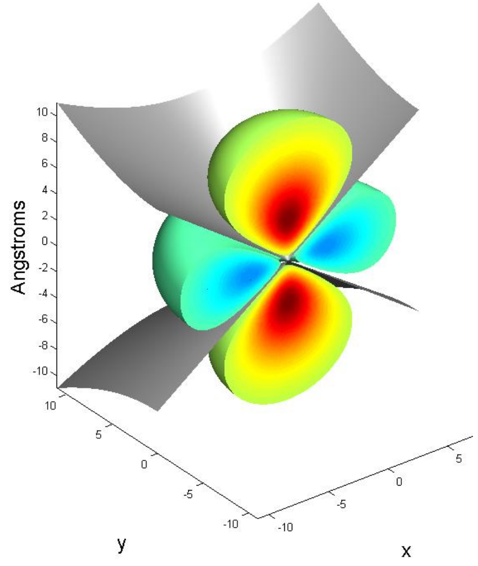
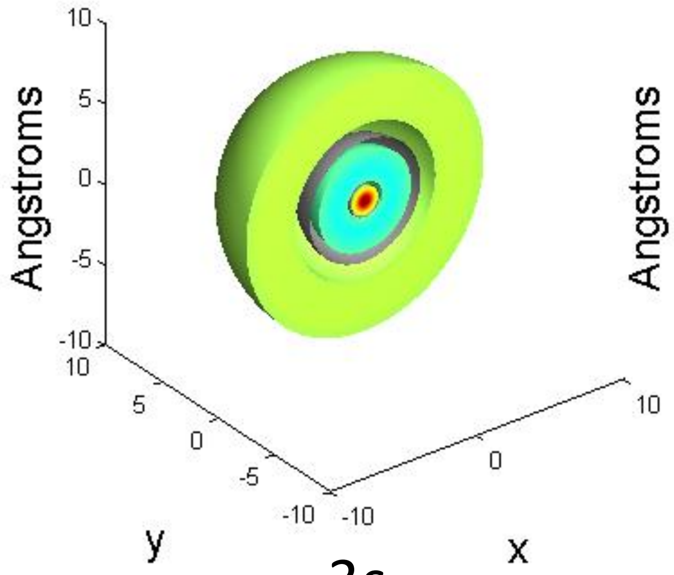


FIGURE 5.13 The shapes of the five 3d orbitals, with phases and nodal surfaces indicated. The "down-axis" view shows the shapes of the first four orbitals (a)–(d) when viewed down the appropriate axis; the specific example shown here is the d_{xy} orbital viewed down the z-axis.

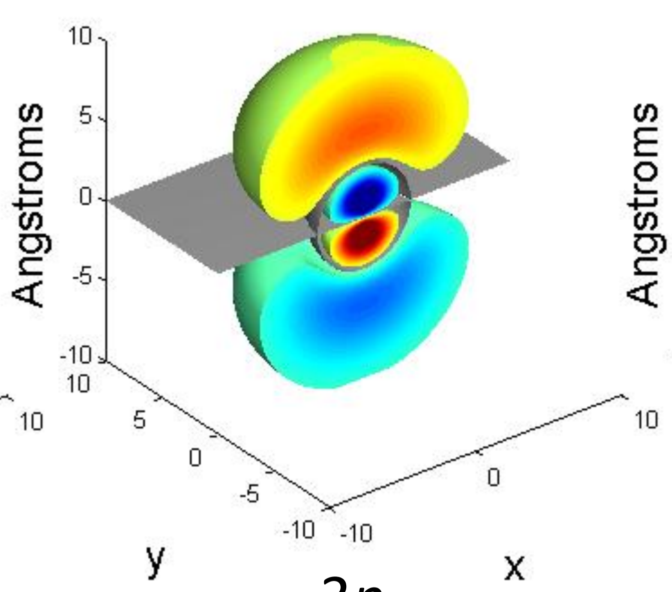


The hydrogen atom: Comparison of 3s, 3p, 3d orbitals



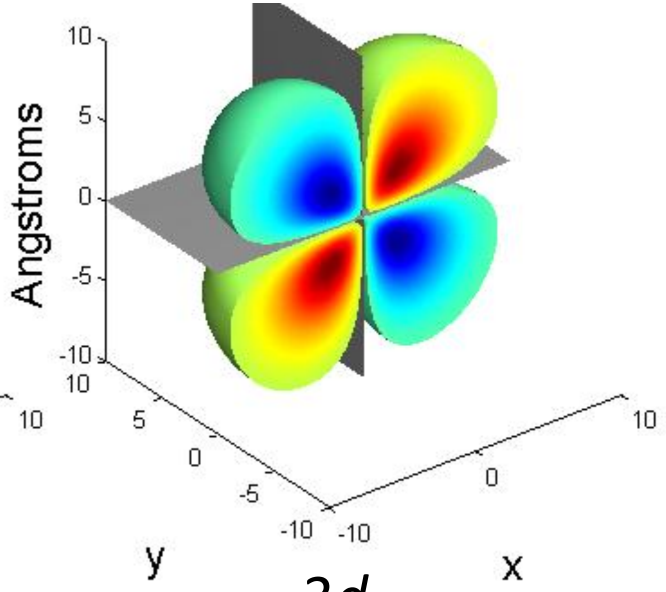
$3s$

n = 3, l = 0.



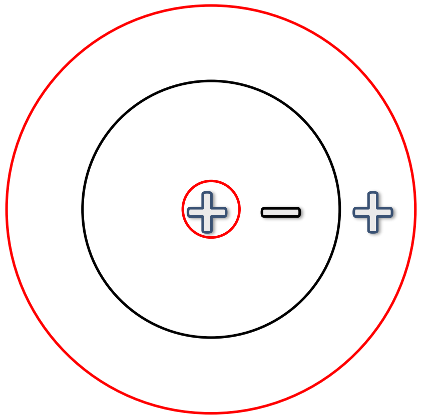
$3p_z$

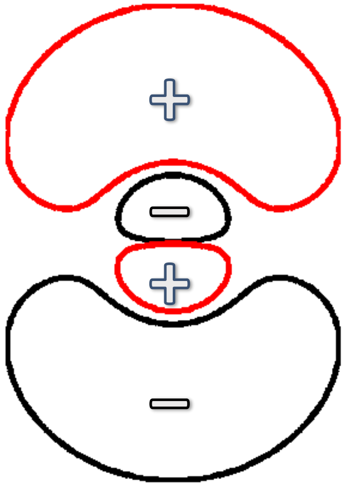
n = 3, l = 1.

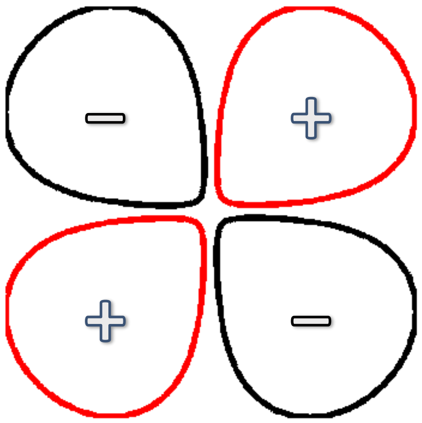


$3d_{xz}$

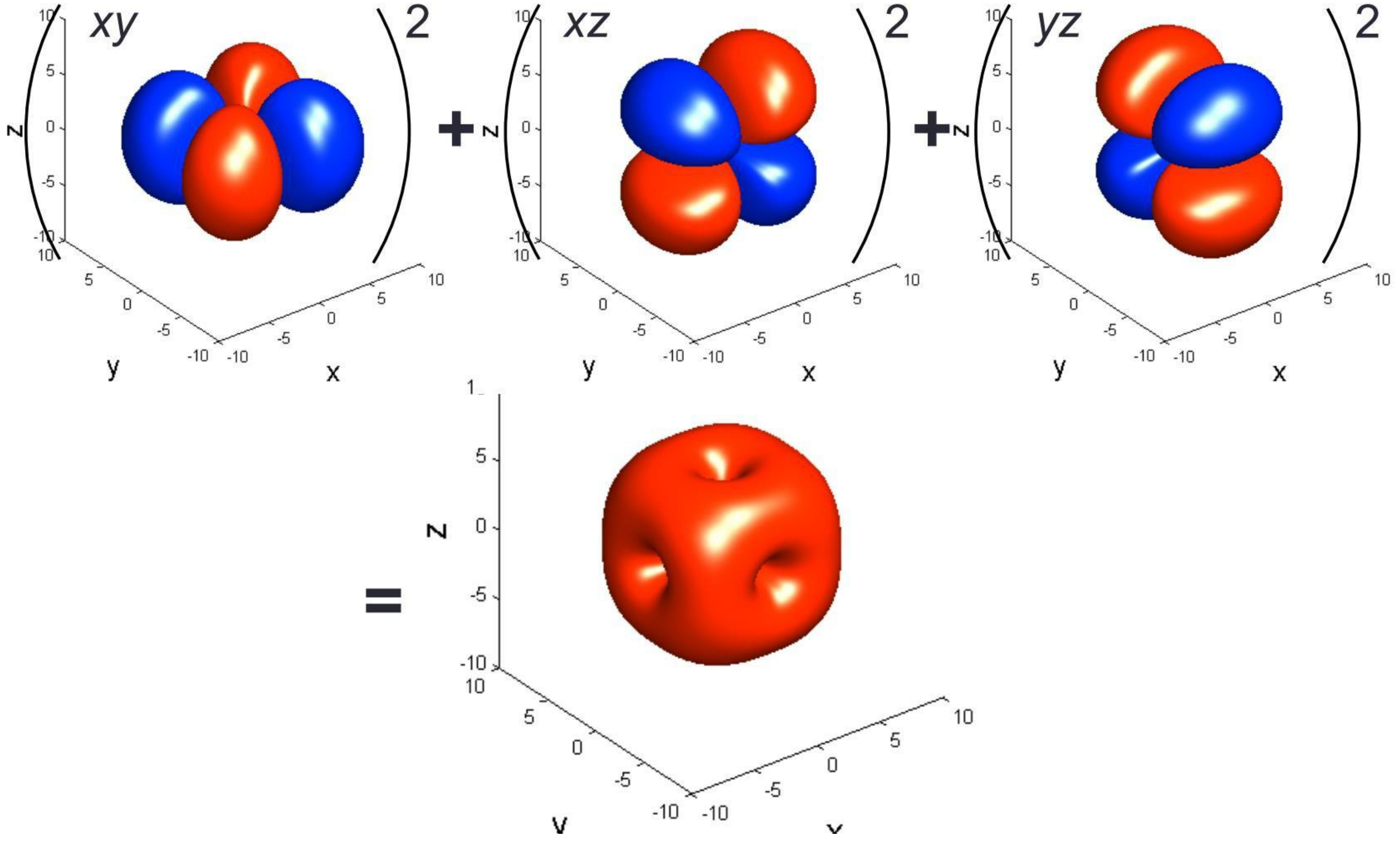
n = 3, l = 2.



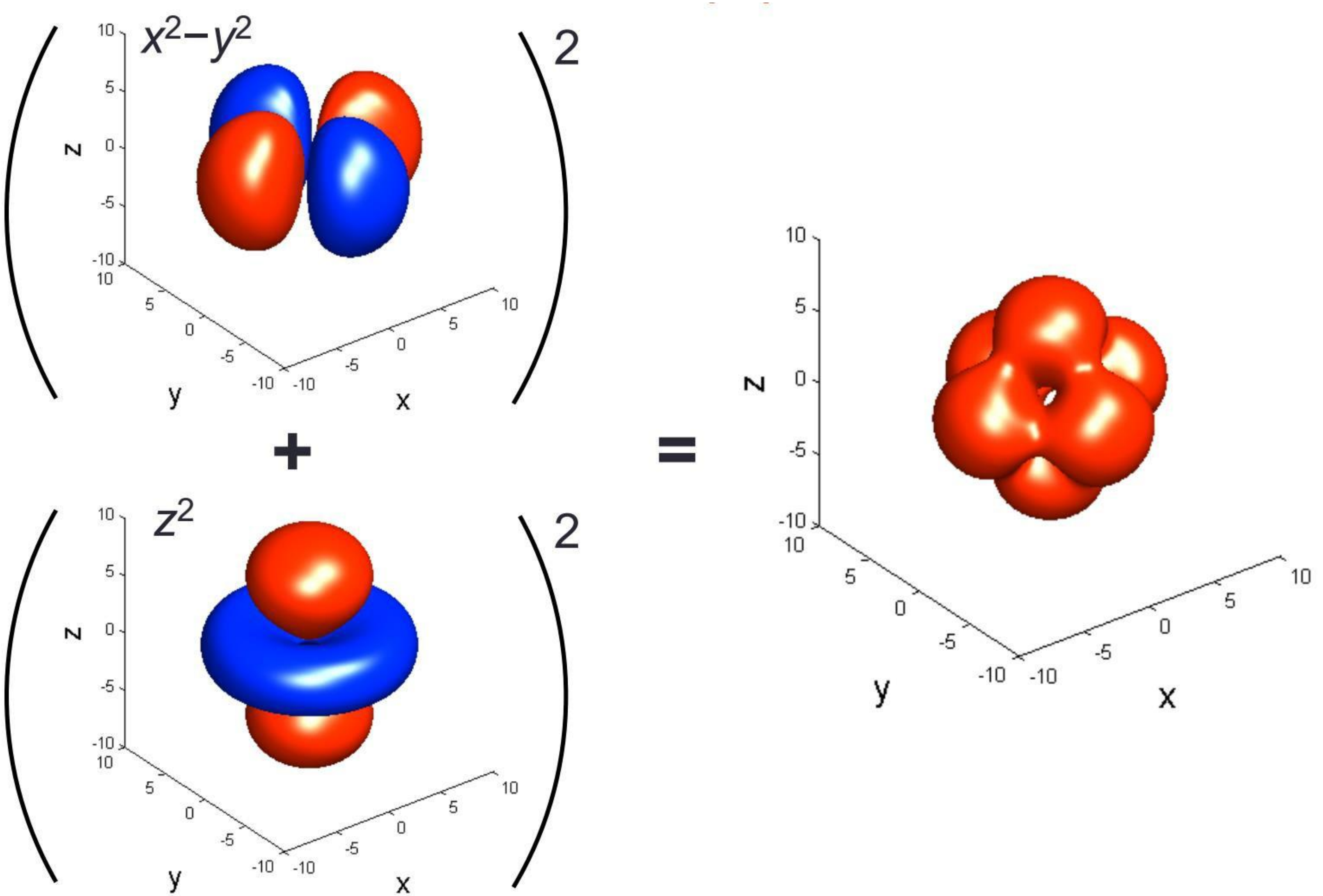




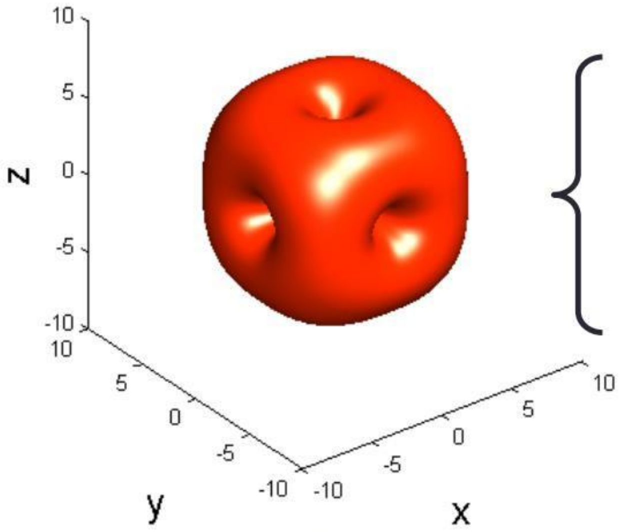
The hydrogen atom: Overlap of d Orbitals 1



The hydrogen atom: Overlap of d Orbitals 2

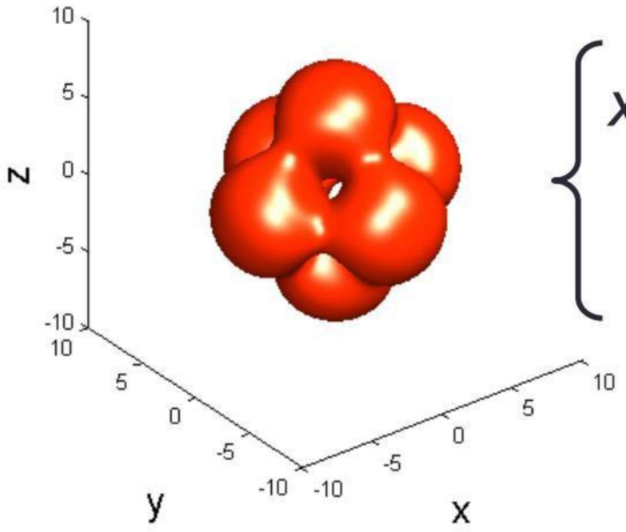


The hydrogen atom: Overlap of d Orbitals 3



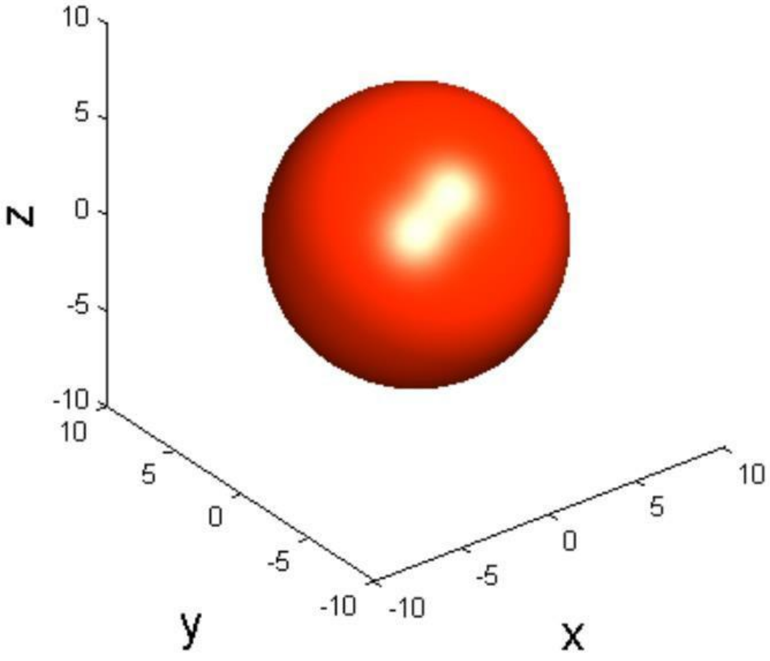
xy
xz
yz

+



x^2-y^2
z^2

=



The five *d* orbitals occupy separate parts of the space.

The hydrogen atom: summary

- For a given ℓ: an increase in n leads to an increase in the average distance of the electron from the nucleus, and the size of the orbital

- An orbital (quantum numbers n and ℓ): ℓ angular nodes and n - ℓ - 1 radial nodes, giving a total of n-1 nodes. An angular node is defined by a plane. A radial node is defined by a spherical surface.

- As r approaches 0, $\psi(r, \theta, \phi)$ vanishes for all orbitals except s orbitals; thus, only an electron in an s orbital can “penetrate to the nucleus,” that is, have a finite probability of being found right at the nucleus.

The hydrogen atom: summary

$$\bar{r}_{n\ell} = \frac{n^2 a_0}{Z}\left\{1 + \frac{1}{2}\left[1 - \frac{\ell(\ell+1)}{n^2}\right]\right\}$$

The average value of the distance of the electron from the nucleus

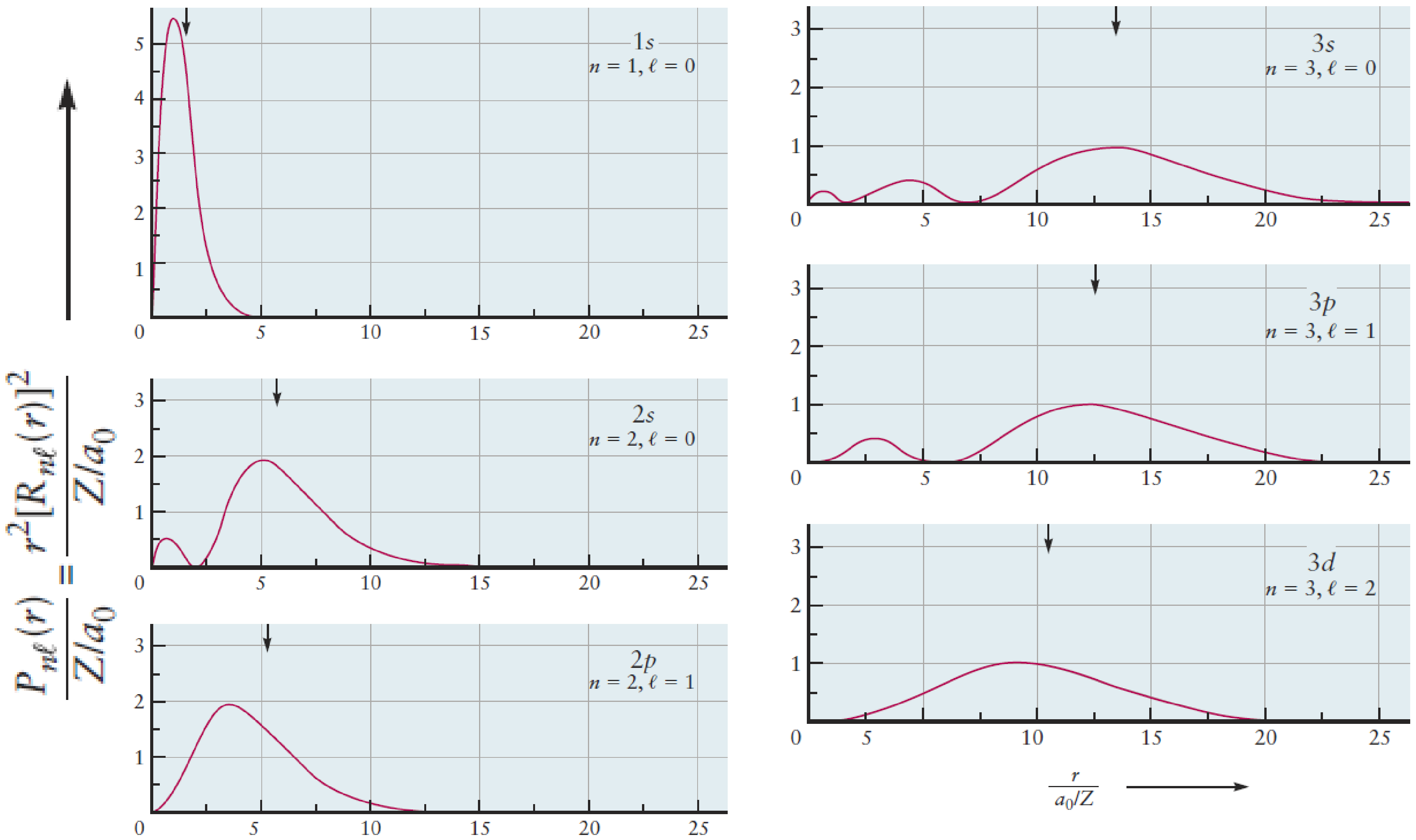


FIGURE 5.14 Dependence of radial probability densities on distance from the nucleus for one-electron orbitals with n = 1, 2, 3. The small arrow above each curve locates the value of $\bar{r}_{n\ell}$ for that orbital. The distance axis is expressed in the same dimensionless variable introduced in Table 5.1. The value 1 on this axis is the first Bohr radius for the hydrogen atom. Because the radial probability density has dimensions (length)$^{-1}$, the calculated values of $r^2[R_{n\ell}(r)]^2$ are divided by $(a_0)^{-1}$ to give a dimensionless variable for the probability density axis.

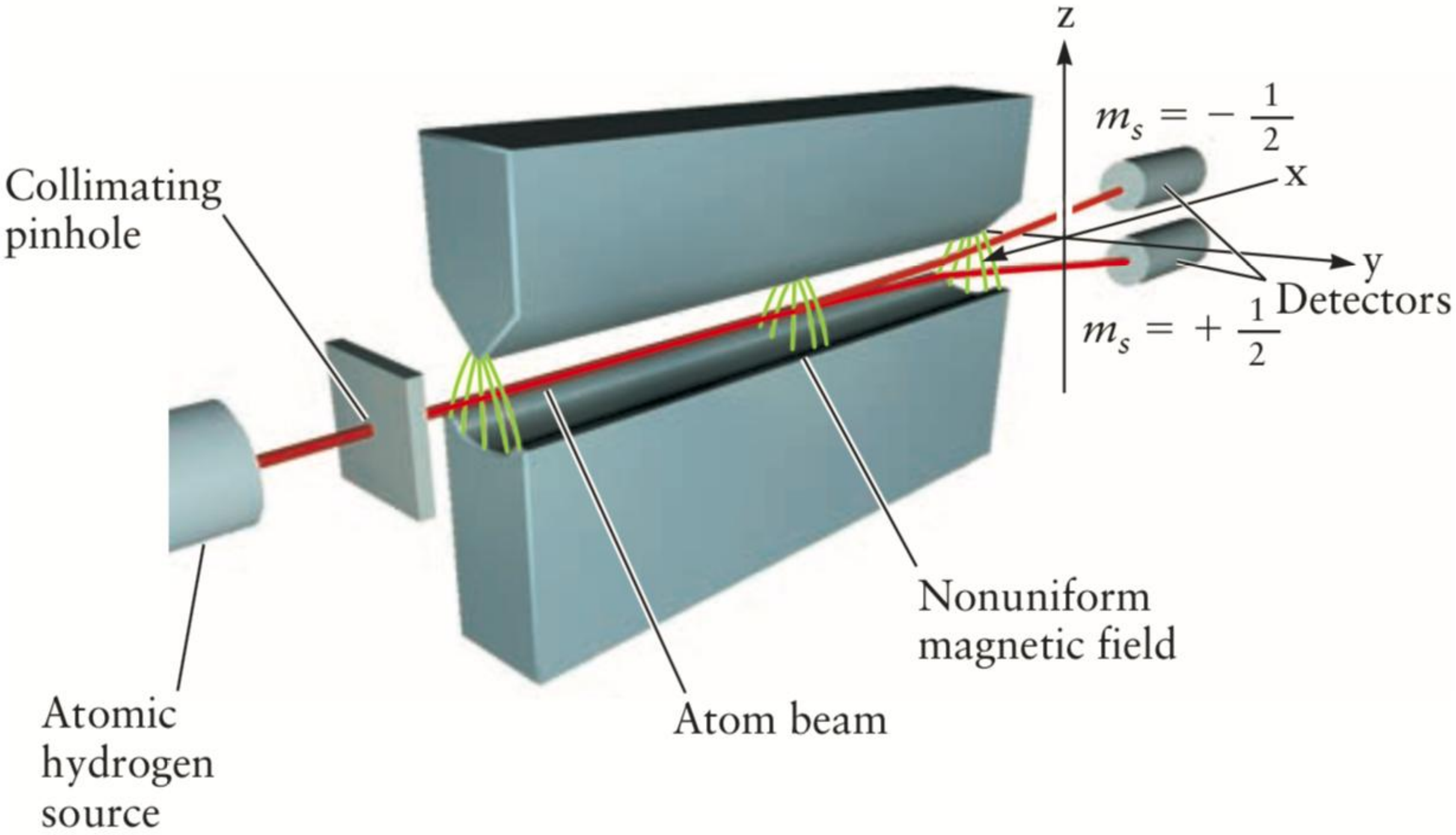
Compare the $3p$ and $4d$ orbitals of a hydrogen atom with respect to the (a) number of radial and angular nodes and (b) energy of the corresponding atom.

Solution

(a) The $3p$ orbital has a total of $n - 1 = 3 - 1 = 2$ nodes. Of these, one is angular ($\ell = 1$) and one is radial. The $4d$ orbital has $4 - 1 = 3$ nodes. Of these, two are angular ($\ell = 2$) and one is radial.

(b) The energy of a one-electron atom depends only on n. The energy of an atom with an electron in a $4d$ orbital is higher than that of an atom with an electron in a $3p$ orbital, because $4 > 3$.

The hydrogen atom: electron spin



That the original beam **is split into only two well-defined beams** in this experiment demonstrates the unexpected fact that **the orientation of the magnetic moment of the electron is quantized**.

The result of this experiment is explained by introducing a fourth quantum number, **m_s**, which can take on two values, conventionally chosen to be **+1/2 and -1/2**. For historical reasons, the fourth quantum number is referred to as the **spin quantum number自旋量子数.**

The hydrogen atom: quantum numbers

$2n^2$ sets of quantum states

$n = 1, 2, 3, \ldots$ 主量子数

$\ell = 0, 1, \ldots, n-1$ 角量子数

$m = -\ell, -\ell+1, \ldots, 0, \ldots, \ell-1, \ell$ 磁量子数

m_s = +1/2 or -1/2 自旋量子数

ℓ = 0 with s, ℓ = 1 with p, ℓ = 2 with d, ℓ = 3 with f, ℓ = 4 with g

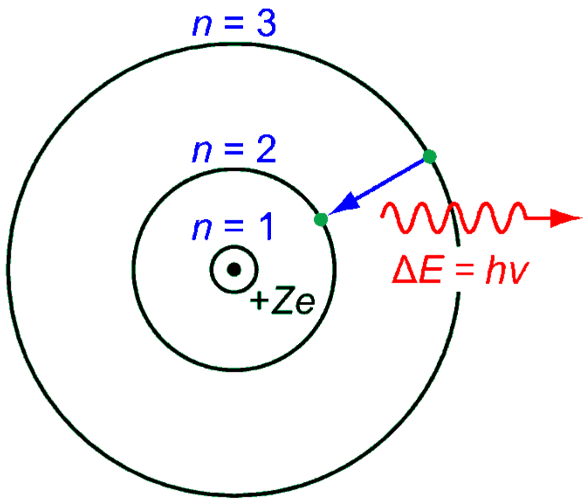
The hydrogen atom: Old vs. New Quantum Theory

Niels Bohr
(1885–1962)

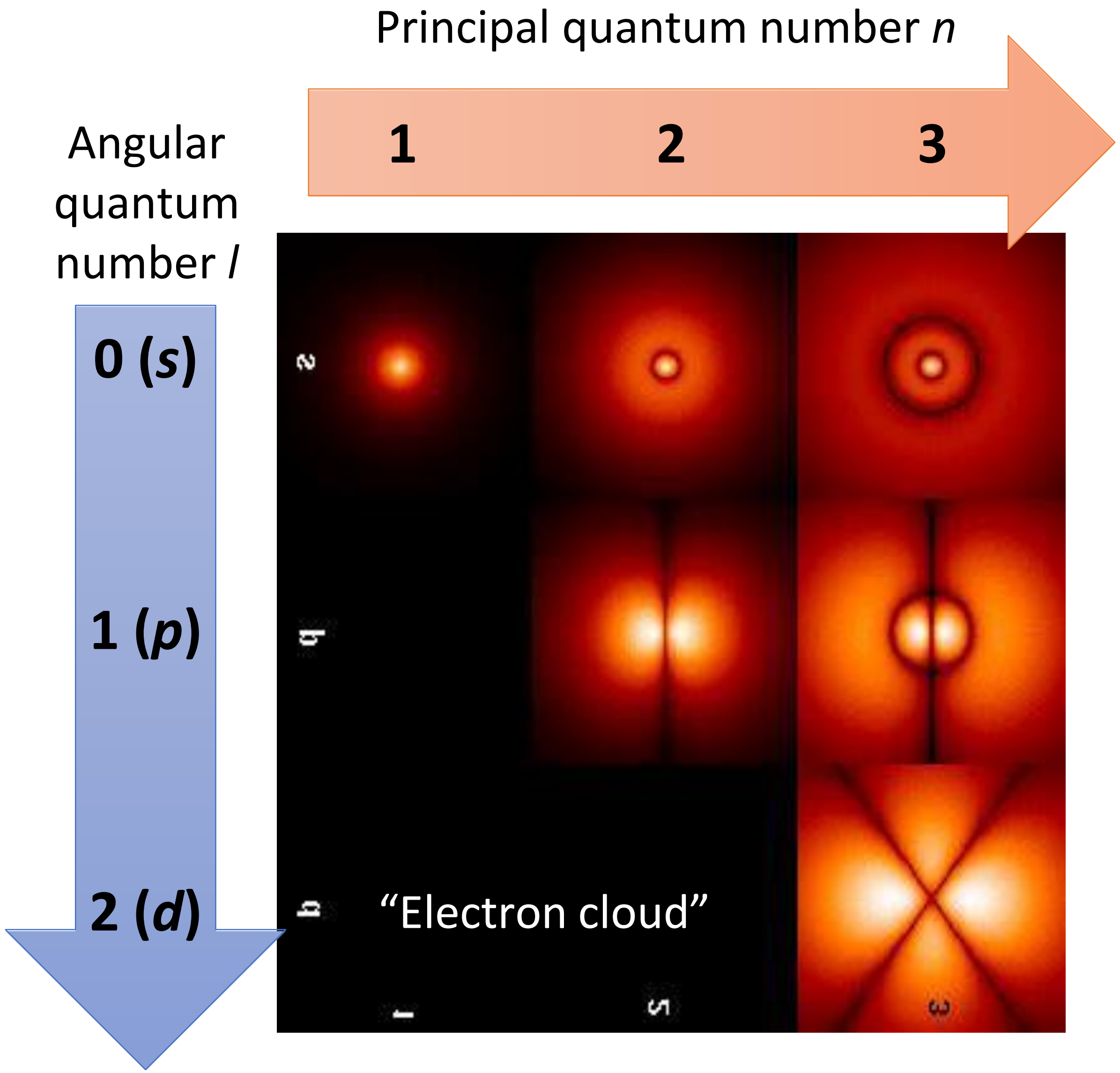
VS

Erwin Schrödinger
(1887–1961)

The hydrogen atom: Orbital Radius



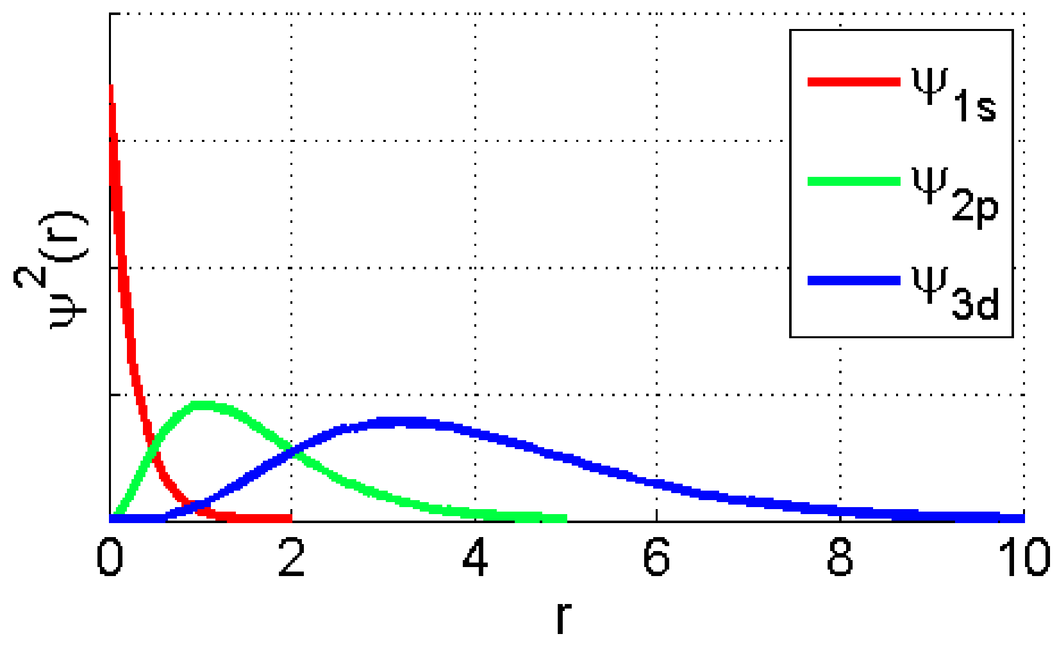
Circular orbits
n = 1,2,3…



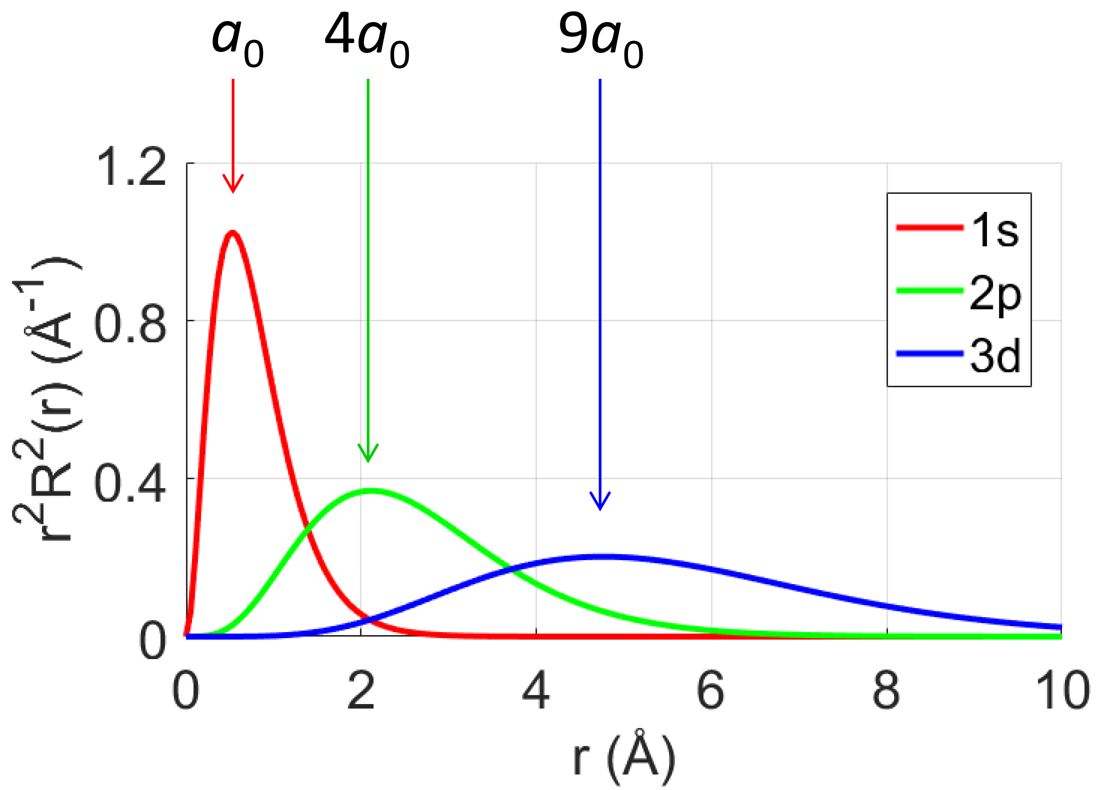
Each p orbital has 3 orientations

Each d orbital has 5 orientations

The hydrogen atom: Orbital Radius



Most probable radius r_{mp}

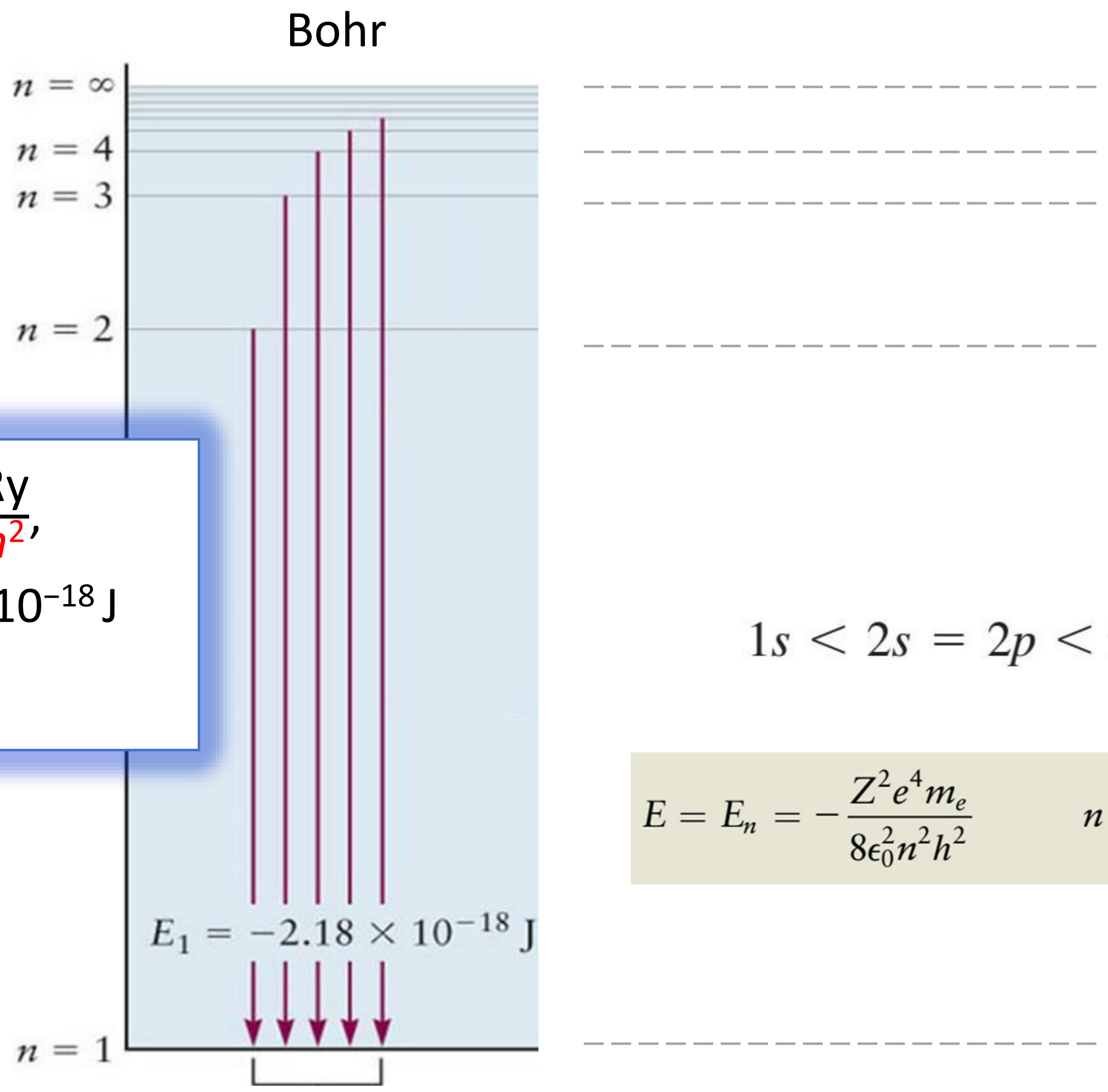


Plot $r^2R^2(r)$ vs. r

For Bohr Model: $r_n = n^2 a_0$

For 1s, 2p, 3d, …: $r_{mp,n} = n^2 a_0$

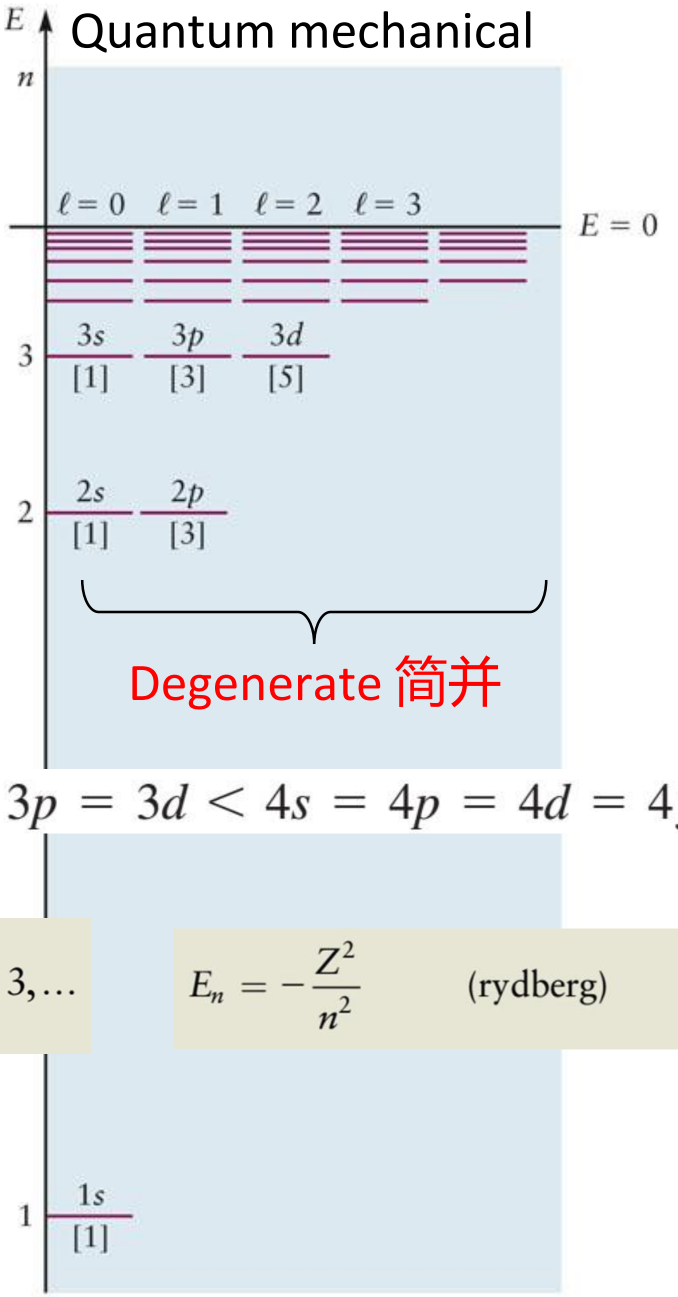
The hydrogen atom: orbital energy



$$E_n = -\frac{\text{Ry}}{n^2},$$

$$\text{Ry} = 2.18\times10^{-18}\ \text{J}$$

$$= 13.6\ \text{eV}$$



$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < \cdots$$

$$E = E_n = -\frac{Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} \qquad n = 1, 2, 3, \ldots$$

$$E_n = -\frac{Z^2}{n^2} \quad \text{(rydberg)} \qquad n = 1, 2, 3, \ldots$$

Homework: read “principle of modern chemistry”, chapter 5.1